

Anatomy of a Design Skill: Which Guidance Improves an LLM’s UI Code, and Where It Acts

(author list withheld for review)

Project 19 — Two-stage study on Qwen2.5-Coder-7B-Instruct

Preprint. Outcome-contingent draft; numeric slots pending the GPU execution phase.

Abstract

Large language models produce visually generic front-end code by default—the “purple-slop” aesthetic—and a short natural-language *design skill* placed in the system prompt is widely reported to fix it. The effect is folklore: everyone reports it, nobody has measured it. We measure it, on one open-weights model (Qwen2.5-Coder-7B-Instruct), in two committed stages that describe the *same* system. **Stage 1** is a black-box factorial ablation of a five-component design skill (color, layout, typography, component patterns, negative constraints), with a length-matched neutral control that isolates design *content* from prompt mass. **Stage 2** is white-box mechanistic interpretability: we locate the skill’s signal in the residual stream, extract a diff-in-means “clean-design” direction, and causally verify it by steering the model with *no design prompt* on held-out prompts, including task types never seen during development. Every quality claim must move *two independent signals*: deterministic objective metrics on the rendered DOM and screenshot, and a style-controlled Bradley–Terry preference from a checklist-anchored VLM judge, gated on chance-corrected human agreement and statistical power; an effect that moves neither is reported as null.

We find that the intact skill improves quality over the length-matched neutral prompt by `NUM:HSKILL-POC` (standardized δ on the primary objective composite) and is preferred `NUM:HSKILL-WIN` of the time; the skill’s causal weight concentrates in `NUM:ABS-NEC-TOP`, and no single component alone reproduces the full effect (sufficiency is distributed). At the representation level, the design signal is `NUM:ABS-LOC` and steering the extracted direction with no design prompt reproduces `NUM:STEER-REPRO` of the skill’s held-out quality gain.

OUTCOME Stage 2 – a causally sufficient design direction exists

A single mid-network direction, added with no design prompt, raises held-out UI quality on both oracle signals, is not reproduced by a norm-matched random control, and—when projected out while the skill is present—attenuates the skill’s own gain; to our knowledge this is the first causal, steerable direction for the *visual quality* of generated front-end code.

OUTCOME Stage 2 – the honest null

No single linear direction steers held-out UI quality under the coherence guardrail: with tight confidence intervals that rule out under-powering, we conclude that—in this model—“clean design” behavior is distributed or not linearly steerable, a precise negative result that bounds the linear-representation account of aesthetic guidance.

The honest-null option is a first-class outcome, not a fallback: it is written up with the same rigor as a positive finding.

1 Introduction

Ask a code model for a landing page and you tend to get the same page: a purple-to-indigo gradient hero, an **Inter** headline over one big statistic, three centered feature cards with rounded corners. Practitioners call it “AI slop” or “the purple gradient website,” and attribute it to models sampling the high-probability center of their web training data—the median of a decade of framework tutorials. The widely reported fix is a short natural-language *design skill*: a few hundred tokens of guidance, dropped into the system prompt, that tells

the model to make deliberate, opinionated choices and to avoid the slop tells. Anthropic ships such a skill, and its plugin reports hundreds of thousands of installs [Anthropic, 2025b,a]; the community credits it widely. Yet the effect has never been measured: which parts of the guidance carry the causal weight, how large the effect is under a defensible quality oracle, and—more fundamentally—*where inside the model* a few sentences of prose act to change what a page looks like.

This paper answers two questions about one system, and insists they describe the same system. **(Q1) Which guidance matters?** We decompose a design skill into five components and run a factorial ablation: leave-one-out for necessity, add-one-in for sufficiency, and a resolution-V fraction for interactions, against a control that holds prompt length constant so the effect cannot be an artifact of prompt mass. **(Q2) Where does it act?** We locate the skill’s signature in the residual stream, extract a diff-in-means direction, and test whether that direction, injected with *no design prompt*, causally reproduces the skill’s quality gain on held-out prompts. Because Stage 2 requires white-box activation access, the subject model is open-weights and self-hosted, and Stage 1 runs on the *same* model (Qwen2.5-Coder-7B-Instruct); otherwise the link between “which instruction matters” and “where it lives” would be severed.

The methodological spine is a two-signal *oracle* we hold to throughout: no quality claim counts unless it moves (i) deterministic objective metrics computed on the rendered DOM and screenshot—never raw code strings—and/or (ii) a style-controlled pairwise preference from a checklist-anchored VLM judge, admitted only after passing a chance-corrected human-agreement gate and a power gate. This is a direct response to two facts from the literature: objective interface-aesthetics metrics explain at most about half the variance in human appeal [Miniukovich and De Angeli, 2015, Reinecke et al., 2013], so the objective half is necessary but insufficient; and (V)LM judges are reliable on UIs when checklist-anchored [Zhang et al., 2025, Jeon et al., 2025] but biased by length and verbosity [Zheng et al., 2023], so the preference half must be style-controlled and human-validated. An effect that moves neither signal is reported as null, and a null is a result.

Contributions. Stated precisely, and credited against the nearest prior work in Section 2:

1. **The first causal component-ablation of a “design skill.”** Existing UI benchmarks score *models or tools* [Si et al., 2024, Zhang et al., 2025, Xiao et al., 2025]; to our knowledge none decomposes the *guidance* and attributes quality to its components with a factorial design and a two-signal oracle. We report which components are necessary, which are sufficient, how they interact, and how a 391-token skill compares to a five-word “make it beautiful” nudge.
2. **The first activation-level interpretability of aesthetic UI-code generation.** Activation steering has been applied to refusal [Arditi et al., 2024], persona and style [Chen et al., 2025, Konen et al., 2024], and broad capabilities [van der Weij et al., 2024]—but, to our knowledge, no prior work extracts or steers a direction for the *visual/aesthetic quality* of generated front-end code. We test the hard, causal bar: does the direction reproduce the gain with no design prompt, on unseen task types, under a coherence guardrail, with a specificity control and a necessity flip test.
3. **A prompt-level↔representation-level correspondence bridge.** We ask whether the components the ablation finds most causal map onto separable steering sub-directions inside the model—do color/layout/typography ablation signatures correspond to near-orthogonal activation directions that each steer their own metric family? This links a black-box explanation to a white-box one, a bridge predicted plausible by the geometry of linear representations [Park et al., 2024a].
4. **The honest null as a first-class outcome.** Every research question carries a preregistered null branch. If no clean steerable direction exists, we report that design behavior is distributed or not linearly steerable in this model, with confidence intervals tight enough to distinguish a real null from under-powering—a legitimate, publishable result, never forced and never buried.

We do not claim to invent diff-in-means extraction, style/persona steering, the two-signal UI oracle, or the objective aesthetics metrics; each is credited in Section 2. The contribution is the synthesis pointed at a new target—design skill → UI code → activations—with the causal rigor (both steering arms, a specificity control, an honest failure surface) the target demands.

Reading map. Section 2 positions the work. Section 3 gives the study design (research questions, the skill and its decomposition, the corpus, the confound controls, preregistration). Section 4 defines the two-signal oracle. Section 5 reports Stage 1 (RQ1–RQ4) with a completed interpretation for every preregistered

outcome. Section 6 gives the Stage 2 method and Section 7—the pivotal section—its outcome-contingent results. Sections 8–10 discuss, bound, and reproduce. Numeric results awaiting the GPU execution phase are shown as `NUM:EXAMPLE` placeholders, each catalogued to the pipeline artifact and RUNBOOK session that fills it.

2 Related work

We situate the study in four literatures and name, in each, the nearest neighbor and the precise gap we fill.

2.1 UI-code generation and its evaluation

A fast-growing line generates front-end code and asks how good it is. Si et al. [2024] (Design2Code) frame screenshot-to-HTML and, crucially for us, evaluate on the *rendered* page with block-level visual metrics rather than raw code—the machinery we adopt in a reference-free form. Wu et al. [2024b] improve UI code via automated feedback with a hard render-success/compilability gate; datasets such as WebSight [Laurençon et al., 2024] and Web2Code [Yun et al., 2024] scale the paired data. On evaluation, ArtifactsBench [Zhang et al., 2025] reports that a checklist-anchored MLLM judge reaches $\sim 94\%$ agreement with human/WebDev-Arena rankings; DesignBench [Xiao et al., 2025], UI-Bench [UI-Bench authors, 2025], Vision2Web [He et al., 2026], and WebDev Arena [LMarena, 2025] score *models or tools*; G-FOCUS [Jeon et al., 2025] studies robust pairwise UI assessment with position-bias control. Accessibility is measurable deterministically via axe-core [Deque Systems, 2024] against WCAG 2.1 [W3C, 2018]. **Gap.** All of these score *a system’s output quality*. None decomposes the *guidance* that produces the output and attributes quality to its components. Our Stage 1 turns the evaluation apparatus of this literature into the response variable of a controlled experiment on the instruction.

2.2 Computational aesthetics of interfaces

Quantifying visual quality has a long history: Ngo et al. [2003] define balance, equilibrium, symmetry, and regularity from element geometry; Miniukovich and De Angeli [2015] extend these to real interfaces and—centrally for our oracle—find they explain $\lesssim 49\%$ of the variance in perceived aesthetics. Reinecke et al. [2013] show colorfulness (a moderate optimum) and visual complexity (an inverted-U) predict first impressions with $\sim 50\%$ ceiling; Hasler and Süssstrunk [2003] give the colorfulness statistic we use. Learned scorers such as UIClip [Wu et al., 2024a] assess UI design data-drivenly. **Gap and consequence.** Because these metrics cap at roughly half the variance, we treat every objective metric as a *diagnostic*, gate learned scorers behind a human-correlation check, and make aesthetic *quality* claims ride on the paired preference signal—the two-signal oracle of Section 4. We also turn the skill’s own negative constraints into a validated “purple-slop index,” making the folk notion of AI slop a computable quantity.

2.3 Prompt- and instruction-component analysis

The methodological shape of Stage 1—compare a behavior with vs. without a piece of the instruction—appears in contrastive-prompt steering [Rimsky et al., 2024, Chen et al., 2025], and the confound it must defeat (adding content also adds tokens) is exactly what Chatbot-Arena style control removes by regressing response length and format out of a Bradley–Terry model [Chiang et al., 2024]. Persona-vector extraction pointedly uses *length- and format-matched* contrastive system prompts [Chen et al., 2025], the analytic twin of our length-matched neutral control. We formalize the ablation as a fractional-factorial experiment [Montgomery, 2017], separating necessity (leave-one-out), sufficiency (add-one-in), and interactions (a resolution-V fraction). **Gap.** We are not aware of prior work applying a factorial design plus a two-signal oracle to the components of a *design* skill for code generation; the synthesis of DOE, confound-matched controls, and a rendered-output oracle is new to this target.

2.4 Activation steering and mechanistic interpretability

Stage 2 builds on the diff-in-means lineage. Ardit et al. [2024] show refusal is mediated by a single residual-stream direction, extracted by difference-in-means and verified by two causal arms—activation *addition* and

directional *ablation* (weight orthogonalization); this is our exact skeleton, with “skill vs. length-matched neutral” replacing “harmful vs. harmless.” Contrastive Activation Addition [Rimsky et al., 2024] and ActAdd [Turner et al., 2023] establish add-a-vector steering; style vectors [Konen et al., 2024] and persona vectors [Chen et al., 2025]—our nearest neighbors—steer *style/trait* using the mean-over-response-tokens statistic we adopt for free-form generation, and Persona Vectors do so on the same model family (Qwen2.5-7B-Instruct), de-risking feasibility. Representation Engineering [Zou et al., 2023] supplies the PCA/LAT cross-check. The reliability caveats are decisive for how we report: steering vectors are heterogeneous, with up to $\sim 50\%$ *anti-steerable* inputs for some concepts [Tan et al., 2024] and in/out-of-distribution fragility [Queiroz Da Silva et al., 2025], so we report per-prompt distributions and an anti-steerable fraction, pre-screen by class separation, and hold out unseen task *types*. On why diff-in-means rather than sparse autoencoders: AxBench [Wu et al., 2025], the largest head-to-head benchmark, finds diff-in-means beats SAE steering and leads concept detection, and there is no public SAE suite for Qwen2.5-Coder [Deng et al., 2026]; steering-vector SAE decompositions can be unfaithful [Mayne et al., 2024]. We therefore adopt diff-in-means as primary and demote SAEs to a labeled stretch. We locate with linear probes plus control-task selectivity [Hewitt and Liang, 2019, Alain and Bengio, 2017] and denoising activation patching [Zhang and Nanda, 2024, Heimersheim and Nanda, 2024]; we interpret through the linear-representation hypothesis and superposition [Park et al., 2024b,a, Elhage et al., 2022], while keeping non-identifiability on the table—a “single direction” may be one face of a concept cone [Wollschläger et al., 2025], and not all features are linear [Engels et al., 2024]. Steering strength is non-monotone [Taimeskhanov et al., 2026], motivating a swept coefficient under a KL guardrail; broad-skill [van der Weij et al., 2024] and conditional [Lee et al., 2025] steering supply our fallbacks. **Gap.** An exhaustive sweep of this literature finds no work extracting or steering a direction for the *visual/aesthetic quality* of generated front-end code. That is the differentiated Stage 2 contribution, and we hold it to the causal bar the reliability literature demands.

3 Study design: problem framing and data

3.1 The subject model, locked

Both stages run on Qwen2.5-Coder-7B-Instruct [Hui et al., 2024] (28 layers, width $d = 3584$). The lock is load-bearing: Stage 2 needs white-box activation access, so the model is open-weights and self-hosted, and Stage 1 uses the *same* model so that “which instruction matters” and “where it lives” describe one system. API models appear only as evaluation judges, never as the subject. Sampling is fixed across every cell and both engines: temperature 0.7, top- p 0.9, top- k 40, repetition penalty 1.05, ≤ 4096 new tokens, seeds $\{0, \dots, 4\}$ (interaction cells $\{0, 1, 2\}$). Temperature 0.7 is deliberate: greedy decoding would collapse seeds to a single output and defeat the distributional analysis.

3.2 Research questions and decision rules

Table 1 states each research question as *hypothesis · endpoints · decision rule · outcomes*, compressed from the preregistration. The two primary endpoints recur throughout: the **Primary Objective Composite** (POC; Section 4) as the objective signal, and a **style-controlled Bradley–Terry utility** as the preference signal. “Reliability-gated” means the judge passed the human-agreement gate; “powered” means the contrast met its McNemar pair count for the observed effect. Every RQ has an explicit honest-null branch (Section 5, 7).

3.3 The independent variable: a five-component design skill

The skill (frozen text in Appendix A) is decomposed into five components, adapted from the Anthropic **frontend-design** skill and community “purple-slop” discourse [Anthropic, 2025b]: **C1** color/palette, **C2** layout/spacing/hierarchy, **C3** typography, **C4** component patterns, **C5** negative constraints. A deliberate design choice makes C1–C4 *purely prescriptive* and consolidates *all* negative constraints in C5, so the components are content-orthogonal: ablating C5 removes all “avoid” guidance, and no “no purple gradients” hides inside the color component. The five components are token-balanced to within $\pm 15\%$ of the ~ 85 -token mean (a build-time assertion using the Qwen tokenizer); the full skill is 391 exact tokens.

Table 1: Research questions, endpoints, and preregistered decision rules (Table T5). Holm controls the family-wise error over the 12-test confirmatory family at 0.05; BH controls FDR at 0.10 for exploratory scans. The full frozen rules are in the preregistration (Appendix F).

RQ	Question / hypothesis	Decision rule	Outcomes
–	H-skill: $\text{FULL} \succ \text{NEUTRAL}$ on both signals (skill is content, not mass)	non-null per oracle; report δ and %-gap over NOSYS	works / null
RQ1	Component <i>necessity</i> : $\mu(\text{FULL}) - \mu(\text{LOO}-C_i) > 0$	C_i necessary if $\delta > 0$ and Holm-sig. on POC or gated BT $\text{CI} > 0$; rank by δ	necessary (ranked) / null / <i>negative</i>
RQ2	Component <i>sufficiency</i> : $\mu(\text{AOI}-C_i) - \mu(\text{NEUTRAL}) > 0, < \text{FULL}$	sufficient-partial if Holm-sig./gated; <i>distributed</i> if none reaches 30% of the gap	one-carries / distributed / null
RQ3	<i>Interactions</i> : some two-factor interaction $\neq 0$	pair interacts if BH-sig. on POC and concordant on ≥ 1 other family	additive / interacting (named)
RQ4	Nudge: $\mu(\text{FULL}) > \mu(\text{BEAUTY1}) > \mu(\text{NEUTRAL})$	content if $\text{FULL}-\text{BEAUTY1} > 0$ gated; nudge-suffices if $\text{BEAUTY1} \geq 70\%$ gap and $\text{FULL}-\text{BEAUTY1}$ null	content / nudge / neither
RQ5	Locate: signal peaks in a contiguous mid-band	band = layers with $\text{AUC} \geq 0.80$, selectivity ≥ 0.15 , patch $\geq 25\%$	localized / diffuse
RQ6	The bar : steer with <i>no</i> design prompt raises held-out quality	four-conjunct success criterion (Section 6)	sufficient (&/or necessary) / null
RQ7	Failure surface: generalizes OOD with bounded anti-steerable fraction	unseen-type $\delta > 0$ gated and anti-steerable $< 50\%$	robust / partial / brittle
RQ8	Correspondence: component subvectors near-orthogonal, each steers its family	mean $ \cos < 0.3$ and $\geq 3/5$ signatures match	found / partial / absent

3.4 The 24-cell factorial and the resolution-V fraction

Encode each cell by a sign vector $x \in \{\pm 1\}^5$ (+1 real component, -1 its matched filler; factor map $A \dots E = C_1 \dots C_5$). We take the 2^{5-1} principal fraction with generator $E=ABCD$, defining relation $I=ABCDE$, which keeps the 16 corners with an even number of -1 's. This fraction is *resolution V*: all five main effects and all ten two-factor interactions are estimable, each aliased only with (assumed-negligible) three- and four-factor interactions (Appendix C). Six of its 16 runs coincide with named cells—the all-+ run is **FULL**, the five single-+ runs are the **AOI** cells—so it contributes 10 new three-component cells. With four controls, five leave-one-out (LOO) cells, and five add-one-in (AOI) cells, the deduplicated design has $4 + 5 + 5 + 10 = \mathbf{24}$ distinct cells (Table 2). LOO and NEUTRAL lie in the *complementary* fraction (odd #minus) and enter as replicated paired contrasts, not as factorial-model rows.

The reason necessity (RQ1) and sufficiency (RQ2) are asked separately is exact. Writing the multilinear factorial model and evaluating at the relevant corners (Appendix C, Eq. 1) gives, for component C_i ,

$$\underbrace{\mathbb{E}[y_{\text{FULL}}] - \mathbb{E}[y_{\text{LOO}-C_i}]}_{\text{necessity Nec}_i} = 2\beta_i + 2\sum_{j \neq i} \beta_{ij} + R, \quad \underbrace{\mathbb{E}[y_{\text{AOI}-C_i}] - \mathbb{E}[y_{\text{NEUTRAL}}]}_{\text{sufficiency Suff}_i} = 2\beta_i - 2\sum_{j \neq i} \beta_{ij} + R, \quad (1)$$

so necessity and sufficiency coincide only when a component's interactions vanish; their average recovers the main effect $2\beta_i$ and their half-difference isolates its total interaction load. Positive synergies ($\sum_j \beta_{ij} > 0$) make a component look *more necessary than sufficient*—components carry the skill together. This single identity ties RQ1, RQ2, and RQ3 and predicts the directional prior that C5 (negatives) helps more once a palette is present.

3.5 Confound controls as methodological contributions

Three controls are engineered, not incidental, and we present them as contributions. (i) *Constant prompt mass*. Every factorial and LOO cell holds 391 exact tokens: a removed component is replaced *in its original slot* by a token-length-matched filler, so cells differ only in *content*, never in length or slot order. NEUTRAL is the all-filler ($-, -, -, -, -$) corner—a length-matched design-irrelevant prompt, which is why a $\text{FULL} \succ \text{NEUTRAL}$ gain is design content and not prompt mass. (ii) *Filler render-inertness*. A filler may touch only source-level conventions (indentation, comments, naming, declaration order) and is forbidden by a build-time audit from mandating anything an objective metric could see—semantic landmarks, `lang/title/meta`, ARIA, validation,

Table 2: The 24 distinct cells (Table T1). + real component, − matched filler, · not a factorial cell. The 16-run factorial is {FULL, AOI-C1..C5, F-123..F-345}; LOO and NEUTRAL are off-fraction contrast anchors. Headline cells get 5 seeds, interaction cells 3.

#	cell	C1	C2	C3	C4	C5	seeds	tier / role
1	FULL	+	+	+	+	+	5	control: intact skill; Stage-2 skill side
2	NEUTRAL	−	−	−	−	−	5	control: length-matched neutral; contrast side
3	BEAUTY1	·	·	·	·	·	5	control: 5-word nudge (RQ4)
4	NOSYS	·	·	·	·	·	5	control: bare base; Stage-2 “no-prompt” base
5–9	LOO-C1..C5	one −, rest +					5	necessity (RQ1)
10–14	AOI-C1..C5	one +, rest −					5	sufficiency (RQ2); in fraction
15–24	F-123..F-345	three +, two −					3	interactions (RQ3); in fraction

or event handlers—so fillers cannot themselves move the oracle and bias component effects downward. (iii) *Constant-mass equalization*. A frozen padding pool deterministically pads or trims each filler until it matches its component’s token count to within ± 2 tokens, recorded in the manifest. A constant output constraint (single-file hermetic HTML) sits in the *user* turn, so it survives even in NOSYS, which has no system message.

3.6 Corpus and the frozen out-of-distribution split

The corpus is 58 realistic briefs across 10 task types, describing domain and function only (full table in Appendix B). A mechanical leakage audit asserts zero whole-word matches of a frozen list of style adjectives and design-family words over the briefs, so the system-prompt treatment is the only design signal in context. The split is frozen by id: **40 dev** (8 development types $\times$ 5), **8 held-out seen-type** (one per development type), and **10 held-out unseen-type**—settings-panel and admin-data-table, two control-/information-dense layouts *entirely absent* from dev. The unseen types are the out-of-distribution generalization test for steering: control-dense layouts unlike the marketing-oriented dev types stress the skill’s guidance in a regime it was not tuned on. A *generation unit* is one (cell, prompt, seed) triple; Stage 1 comprises 4,000 dev units (all 24 cells $\times$ 40) plus 630 held-out units (reduced set), totaling 4,630.

3.7 Preregistration

The cell table, the 58-brief split, the POC formula and its two conditionals, the confirmatory 12-test family, the reliability and power gates, the Stage-2 success criterion, and the null rule are frozen in a preregistration (Appendix F) that is git-tagged before any GPU run. A pilot gate (2 prompts $\times$ {FULL, NOSYS} $\times$ 3 seeds) must show a visible skill effect before the generation budget is spent; the sole sanctioned model-swap trigger is a pilot that still shows no effect after prompt-format fixes.

4 The two-signal oracle

Every quality claim—an ablation delta or a steering delta—must move two independent signals. This section defines them, the null rule that binds them, and the power that backs them. The oracle is not a scoring convenience; it is the paper’s evidentiary standard, and it is preregistered.

4.1 Objective signal: deterministic metrics on the rendered output

All objective metrics are computed on the *rendered* DOM and screenshot from a pinned, offline, animation-disabled headless Chromium at 1440×900 —never on raw code strings—so they are deterministic and unfoolable. They are reported as a *vector* in six families: **V** validity/hermeticity (render success, external-fetch count, overflow and overlap counts, axe-core violations), **A** accessibility/contrast, **L** layout, **T** typography, **C** color, **X** complexity. Two archetypes fix the flavor. Per text node we compute the WCAG contrast ratio between foreground luminance L_1 and background L_2 ,

$$\text{CR} = \frac{L_1 + 0.05}{L_2 + 0.05} \in [1, 21], \quad L = 0.2126 R_{\text{lin}} + 0.7152 G_{\text{lin}} + 0.0722 B_{\text{lin}}, \quad (2)$$

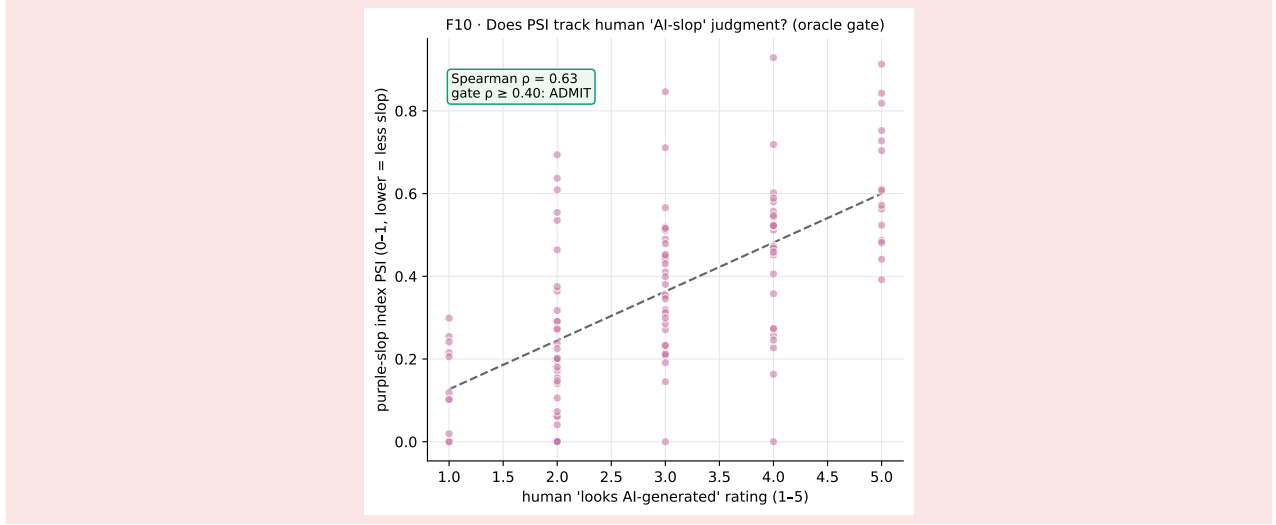


Figure 1: **[F10 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 4).* Does PSI track human “AI-slop” judgment? The admission gate ($\rho \geq 0.4$) decides whether the confirmatory POC keeps the $-PSI$ term (7-term) or drops it (6-term). Axes and the gate annotation are the final layout; the scatter is planted.

(sRGB-linearized channels), reporting the fraction of nodes below 4.5:1. And the skill’s own C5 constraints become a validated **purple-slop index**

$$PSI = 0.30 \text{ hueFrac}[260^\circ, 290^\circ] + 0.30 \text{ gradientBg} + 0.25 \text{ InterRobotoShare} + 0.15 \text{ centeredHero}, \quad (3)$$

each subcomponent in $[0, 1]$; the skill tells us exactly what “slop” is, so we measure exactly that. Layout uses Ngo balance/equilibrium/symmetry and an alignment-regularity count; typography scores modular-scale adherence; color uses Hasler colorfulness M (Appendix C).

The Primary Objective Composite. To have *one* confirmatory objective number per generation while honoring “do not aggregate raw metrics,” we preregister a mean of oriented, z -scored core metrics,

$$POC(\mathbf{x}) = \frac{1}{7} [z(\text{align_reg}) + z(\text{whitespace}) + z(\text{type_scale}) + z(\text{contrast_pass}) - z(\text{overflow_overlap}) - z(PSI) - z(|M - c^*|)] \quad (4)$$

$z(\cdot)$ over all dev generation units, oriented so higher is better; the colorfulness term is a distance to a moderate-optimum target c^* , not “more is better.” Render-failed units are excluded from POC (a non-rendering page is a validity null, counted in family V). Two conditionals are frozen *with* the endpoint, not applied post hoc: (i) if PSI fails its admission gate (below) the POC drops the $-z(PSI)$ term and every confirmatory decision uses the 6-term variant; (ii) c^* is the median M of human-preferred pages if ≥ 30 exist, else the median M of dev FULL-cell generations. The branch that fires is written to the manifest *before* the first confirmatory test.

Oracle validation gates. Before any metric is used as evidence: PSI is admitted only if Spearman $\rho(PSI, \text{human “AI-generated” rating}) \geq 0.4$; on the human subset we observe $\rho = \text{NUM:PSI-RHO}$ (Figure 1), so the **NUM:POCVAR** POC is used and $c^* = \text{NUM:CSTAR}$. The learned UIClip scorer is admitted as *auxiliary* only if $\rho(\text{UIClip}, \text{human win-rate}) \geq 0.3$ (observed **NUM:UICLIP-RHO**); it never enters the POC or a confirmatory decision, because a learned scorer is not deterministically auditable and carries a mobile-distribution shift.

4.2 Preference signal: style-controlled Bradley–Terry with a validated judge

The second signal is a pairwise preference from a checklist-anchored VLM judge (primary: Gemini 2.5 Flash, a different family from Qwen to avoid self-enhancement; GPT-4o audits 15%). Each pair is judged in *both*

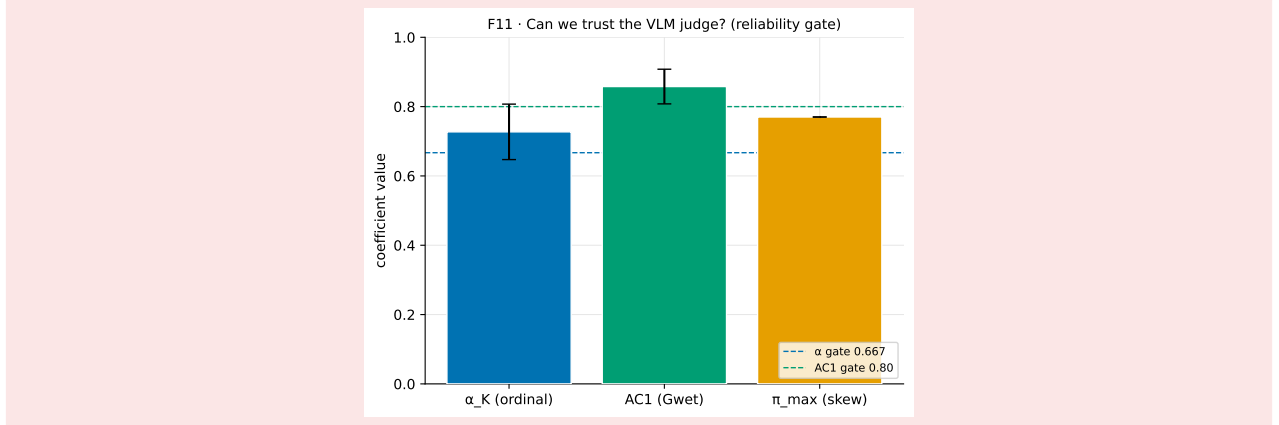


Figure 2: **[F11 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 5).* Can we trust the VLM judge? α_K , AC1, and the marginal skew $\pi_{\max}$ with bootstrap CIs against the two gate thresholds. The disjunctive gate admits AC1 only under demonstrable skew; the bars are the kappa-paradox exemplar (planted), the final panel plots the real judge–human numbers.

presentation orders and counts as a winner only if the verdict is order-consistent, else a tie—a position-bias control. We fit a **style-controlled Bradley–Terry** model with a Davidson tie term,

$$P(i \succ j) = \sigma((\beta_i - \beta_j) + \gamma^\top(s_i - s_j)), \quad s = (\log \text{DOM}, \log \text{tok}, \# \text{elem}, \text{colorfulness}, \text{text-density}), \quad (5)$$

reference $\beta_{\text{NEUTRAL}} = 0$, cluster-bootstrap CIs over prompts ($B=2000$). The covariates are judge-bias channels a designer would never score by (node count, token length); omitting them biases the estimated skill effect by $\gamma^\top(\bar{s}_{\text{FULL}} - \bar{s}_{\text{NEUTRAL}})$, the analysis-level twin of the design-level length-matched control. One covariate is flagged as partially causal: colorfulness *is* a design lever, so regressing it out could absorb a genuine C1 effect. We therefore report both the controlled and uncontrolled $\hat{\beta}$, recover the legitimate color effect through the objective channel (the POC $|M - c^*|$ term and the RQ8 color-family steering signature, neither routed through the judge), and if the two $\hat{\beta}$ disagree in sign we report the contrast as *style-confounded* rather than resolved.

The reliability gate and the kappa paradox. The preference signal counts only if the judge tracks a human. Sean labels $N=100$ stratified pairs in both orders; we compute judge–human Krippendorff α_K (ordinal) and Gwet AC1 with bootstrap CIs. Because the skill usually wins, the marginal is skewed, and a skewed marginal crushes α_K even at high raw agreement—the kappa paradox. The two coefficients differ only in their chance model,

$$\alpha_K = 1 - \frac{D_o}{D_e}, \quad \text{AC1} = \frac{p_o - p_e^\gamma}{1 - p_e^\gamma}, \quad p_e^\gamma = \frac{1}{q-1} \sum_k \pi_k(1 - \pi_k), \quad (6)$$

and $p_e^\gamma \rightarrow 0$ as one category dominates, so AC1 stays near p_o where α_K collapses (a worked example: at 90% agreement, skew moves α_K from 0.80 to 0.44 while AC1 holds at 0.88; Appendix C). The preregistered gate is therefore *disjunctive*:

$$\text{pass} \iff \alpha_K \geq 0.667 \text{ or } (\text{AC1} \geq 0.80 \text{ under demonstrable skew, } \pi_{\max} \gtrsim 0.80).$$

The skew qualifier stops AC1 from being a loophole in the balanced case (where the two agree). We observe $\alpha_K = \text{NUM:REL-ALPHA}$, $\text{AC1} = \text{NUM:REL-AC1}$, $\pi_{\max} = \text{NUM:REL-PIMAX}$ (Figure 2); the gate is **NUM:REL-GATE**. These thresholds are convention, not derived optima, and we report the CIs so a reader may apply a stricter bar.

4.3 The null rule (verbatim) and power

The two signals are bound by a rule we quote exactly as preregistered:

An ablated component’s effect, or a steering direction’s effect, is reported as null unless it moves an objective-metric endpoint (POC Holm-significant, or ≥ 2 concordant BH-significant families) OR a preference delta that passes both the reliability gate and the power gate. Aesthetic claims not backed by the objective half are inadmissible.

The disjunction is a deliberate, bounded loosening for the necessity/sufficiency *screen*: with each channel controlled near 0.05, the false-non-null rate is $\lesssim 0.10$ by a union bound, which is safe because the dominant risk is a *false null* (missing a real component) and the objective channel is deterministic. The Stage-2 headline instead uses a conjunction of four checks, whose joint false-positive rate falls below any single α (Section 6).

Preference deltas are paired, so we size them with McNemar’s test. Among decisive (order-consistent) pairs the winner is tested against a fair coin; the required decisive count at alternative split θ_1 is

$$N_{\text{dec}} \approx \frac{(z_{1-\alpha/2}\sqrt{\theta_0(1-\theta_0)} + z_{1-\beta}\sqrt{\theta_1(1-\theta_1)})^2}{(\theta_1 - \theta_0)^2}, \quad \theta_0 = \frac{1}{2}, \quad (7)$$

giving ≈ 200 decisive pairs for a 60/40 split, ≈ 80 for 65/35, ≈ 780 for 55/45 (at $\alpha=0.05$, power 0.80). The design allocates 200 pairs to FULL-NEUTRAL and ~ 120 to each necessity edge accordingly. Small expected effects (some AOI cells) are lightly powered on *preference* on purpose—their non-null status can still be established through the fully powered objective channel, exactly what the two-signal rule permits. The power gate is an admissibility screen, not an inference: a contrast counts only if it achieved the decisive pairs Eq. 7 requires *at the observed* split; an under-powered delta is never reinterpreted as a null. We observe a tie rate of NUM:POW-TIE, NUM:POW-FN decisive pairs on FULL-NEUTRAL, NUM:POW-NEC per necessity edge, and NUM:POW-UNDER under-powered confirmatory edges (Table T10, Appendix E).

5 Stage 1 results: which guidance matters

All Stage-1 objective deltas are contrasts on the POC via a mixed-effects model $y_{cps} = \mu + \tau_c + u_p + v_{cp} + \epsilon_{cps}$ with fixed cell effect τ_c , random prompt intercept u_p , and random cell-by-prompt slope v_{cp} ; this is what separates the study from per-generation *t*-tests, because five seeds on one prompt are not five independent points—with a modest intraclass correlation the effective n is roughly halved (Appendix C). Preference deltas are style-controlled Bradley–Terry utilities (Eq. 5). Confirmatory contrasts are Holm-corrected over the 12-test family, separately on each signal; the per-component estimates are collected in Table 3 and visualized in Figure 3. For each research question we give the completed interpretation the paper will carry under *every* preregistered outcome; on real data, exactly one branch per question is kept.

5.1 Does the skill work, and is it content?

The headline contrast is FULL vs. NEUTRAL. Because NEUTRAL is length-matched, a gain is design *content*, not prompt mass. We measure NUM:HSKILL-POC standardized δ on the POC and a FULL win-rate of NUM:HSKILL-WIN with BT utility NUM:HSKILL-BT; the prompt-mass control NEUTRAL vs. NOSYS moves the POC by NUM:NEU-NOSYS, and FULL closes NUM:HSKILL-NOSYS-PCT of the NOSYS gap.

OUTCOME works

The design skill improves UI quality on both oracle signals: FULL > NEUTRAL at NUM:HSKILL-POC POC and NUM:HSKILL-WIN preference, both surviving Holm. Because the gain is measured against a token-matched, design-irrelevant prompt, it is attributable to the skill’s *content* rather than to added prompt mass—the near-zero NEUTRAL-NOSYS contrast (NUM:NEU-NOSYS) confirms that filler tokens alone do essentially nothing. The rest of Stage 1 is well-posed: there is a real effect to decompose.

OUTCOME null

FULL $\approx$ NEUTRAL on both signals with a tight CI around zero: the model cannot make use of a structured design skill at all. This outcome would have been caught at the pilot gate and escalated to the recorded model fallback; were it to survive to the full run, we would report it honestly and scope the remainder of Stage 1 to “no measurable skill effect on this model,” a negative result about the model’s instruction-following rather than about design guidance in general.

5.2 RQ1 — component necessity

Necessity is the paired contrast FULL vs. LOO- C_i : does removing C_i from the intact skill degrade quality? Per-component objective deltas are in Table 3 and the twin-panel forest plot (Figure 3); the directional prior is that C5 (negatives) and C1 (color) punch above their token weight.

OUTCOME necessary (ranked)

Removing certain components significantly degrades quality; ranked by standardized δ , the skill's weight concentrates in NUM:NEC-RANK, with NUM:NEC-NSURV of five components surviving Holm on the POC. The largest necessity effects are C5 (negatives, NUM:NEC-C5; BT utility NUM:NEC-C5-BT) and C1 (color, NUM:NEC-C1; BT NUM:NEC-C1-BT), confirming the prior that the skill's leverage lies disproportionately in *telling the model what not to do* and *pinning a palette*—precisely the components whose token weight is smallest relative to their effect. The ordering replicates on held-out prompts (top component NUM:NEC-REPL-TOP), so it is not a dev artifact. Per Eq. 1, components with large necessity but smaller sufficiency (below) carry positive interaction load—they help most in the presence of the others.

OUTCOME null

No leave-one-out contrast degrades quality: every FULL-LOO- C_i CI brackets zero (Table 3), and the power gate certifies these as tight nulls, not under-powering. The reading is that no single component is individually necessary—the skill works only as a whole, with each component substitutable by the others' combined effect. This makes RQ2/RQ3 the load-bearing questions: a skill that is robust to any single deletion is either highly redundant or purely interactive.

OUTCOME negative

Removing a component *improves* quality: at least one FULL-LOO- C_i contrast is significantly *negative* (LOO beats FULL). This is a genuine, honestly reported finding—an over-constraining instruction that, in this model, does more harm than good (a plausible candidate is an over-specified typography or pattern directive that pushes the model into awkward choices). The practical lesson for skill design is that more guidance is not monotonically better; we report which component and by how much, and recommend its revision.

5.3 RQ2 — component sufficiency

Sufficiency is AOI- C_i vs. NEUTRAL: does C_i *alone* lift quality, and does any single component reach FULL? The necessity $\times$ sufficiency scatter (Figure 4) reads the two questions jointly against the $y=x$ line where interactions vanish. The largest single-component lift reaches NUM:SUFF-MAXPCT of the FULL-NEUTRAL gap.

OUTCOME distributed

Every component alone lifts quality above NEUTRAL (NUM:SUFF-NSURV of five Holm-significant), but none reaches the preregistered 30% of the FULL-NEUTRAL gap—the largest, NUM:SUFF-TOP, attains only NUM:SUFF-MAXPCT. Sufficiency is therefore *distributed*: no single instruction carries the skill, and the full effect emerges from the components together. Read with RQ1 through Eq. 1, components that are more necessary than sufficient (above the $y=x$ line in Figure 4) reveal positive synergy—the whole exceeds the sum of parts.

OUTCOME one-carries

A single component reaches within 30% of FULL on its own: NUM:SUFF-TOP attains NUM:SUFF-MAXPCT of the gap, and no other component approaches it. The skill is, to first order, that one instruction; the remainder is polish. This is a strong, compressible result—most of a 391-token skill's benefit is reproducible with one component—and it sharpens the Stage-2 target: the extracted direction should correspond most closely to that component's activation signature.

OUTCOME null

No component alone beats NEUTRAL (all AOI- C_i CIs bracket zero) even though the intact skill does (H-skill). Components are effective only in combination: sufficiency is strictly emergent. This pairs with the RQ3 interaction story—if no main effect is individually sufficient yet the skill works, the effect lives in the interactions, and the factorial block is where it is found.

5.4 RQ3 — interactions and additivity

The 16-run factorial estimates all ten two-factor interactions cleanly (each aliased only with an assumed-negligible three-factor term). The 2FI heatmap is Figure 5; the directional prior is a positive C1 $\times$ C5 (negatives

help more once a palette is set) and a mild negative C2×C3.

OUTCOME interacting

NUM:INT-NSURV two-factor interactions survive BH and are concordant on a second metric family. The largest is **NUM:INT-MAXPAIR** (**NUM:INT-MAX**); as predicted, C1×C5 is super-additive (**NUM:INT-C1C5**)—negative constraints deliver more once a deliberate palette is specified, because “avoid purple gradients” bites hardest when a replacement palette exists—while C2×C3 is mildly sub-additive (**NUM:INT-C2C3**). The components are not independent knobs; the skill is a small interacting system, and this is the quantitative content of the necessity>sufficiency asymmetry in Eq. 1.

OUTCOME additive

No two-factor interaction survives BH: the component effects simply add, and the skill behaves as a *checklist* whose value is the sum of its independent items. This is itself informative for skill design—components can be added, removed, or reordered without surprise—and it means necessity and sufficiency coincide (Eq. 1 with $\sum_j \beta_{ij} \approx 0$), which Figure 4 would show as points on the $y=x$ line.

OUTCOME dominated by a single main effect

One component’s main effect swamps all interactions and most other mains: the skill is effectively one lever (**NUM:SUFF-TOP**), with the remaining components contributing little independently or interactively. The paper’s Stage-1 story then collapses to “the skill is essentially component X ,” and Stage 2 asks whether X ’s activation signature is the whole design direction.

5.5 RQ4 — do five words match 391 tokens?

We compare the intact skill to a five-word nudge (BEAUTY1: “Make it beautiful and well-designed.”) and to NEUTRAL, testing the ordered contrasts. BEAUTY1 reaches **NUM:NUDGE-BPCT** of the FULL–NEUTRAL gap.

OUTCOME content

The ordering FULL>BEAUTY1>NEUTRAL holds with FULL–BEAUTY1>0 (**NUM:NUDGE-FB** POC; FULL preferred over BEAUTY1 at **NUM:NUDGE-FBWIN**): a nudge helps (**NUM:NUDGE-BN** over NEUTRAL) but structured guidance is load-bearing and not substitutable by a one-liner. What the 391 tokens buy beyond “make it beautiful” is the *specific* content—the palette discipline and the negative constraints RQ1 finds most necessary—which a generic exhortation cannot supply.

OUTCOME nudge

BEAUTY1 reaches $\geq 70\%$ of the gap (**NUM:NUDGE-BPCT**) and FULL–BEAUTY1 is null: a five-word nudge captures most of the skill’s effect. This is a striking, cheap result—the model already “knows” how to make a better page and mostly needs *permission* to leave the high-probability center, not 391 tokens of instruction. It reframes the design skill as an activation trigger rather than a knowledge transfer, which is exactly the hypothesis Stage 2 tests mechanistically.

OUTCOME neither

BEAUTY1 $\approx$ NEUTRAL: a vague nudge does nothing, and only the specific structured skill helps (if H-skill fired). The model needs concrete, itemized guidance or none at all—consistent with a picture in which “beautiful” is too diffuse a target to move the output distribution, whereas named colors, type scales, and prohibitions are actionable.

5.6 RQ7 (black-box slice) — does the skill effect generalize across task types?

Before any steering, we report the FULL–NEUTRAL POC delta *per task type*, including the two held-out unseen types (settings-panel, admin-data-table). The mean seen-type delta is **NUM:BB-SEEN** and the mean unseen-type delta is **NUM:BB-UNSEEN**. A skill effect that holds on the information-dense unseen types is evidence the guidance encodes transferable design content, not a marketing-page idiom; a collapse on those types localizes the skill’s benefit to the layouts it was written for and sets expectations for the Stage-2 out-of-distribution steering test.

6 Stage 2 method: locate, extract, steer, verify

Stage 2 asks where the skill acts and whether that locus is causal. All activation work runs on Hugging Face **transformers** with a custom hook module (vLLM cannot inject mid-decode); the residual stream is the layer-output hidden state, and every comparison is within-engine—we never compare a vLLM baseline to an

Table 3: Stage-1 per-component effects on the POC (standardized δ , 95% cluster-bootstrap CI), necessity (FULL-LOO- C_i) and sufficiency (AOI- C_i -NEUTRAL). Interactions and preference utilities are reported in Figures 3–5. Every entry awaits the GPU run.

component	necessity δ (FULL-LOO)	sufficiency δ (AOI-NEUTRAL)	prior
C1 color	NUM: NEC-C1	NUM: SUFF-C1	large
C2 layout	NUM: NEC-C2	NUM: SUFF-C2	medium
C3 typography	NUM: NEC-C3	NUM: SUFF-C3	medium
C4 patterns	NUM: NEC-C4	NUM: SUFF-C4	near-null
C5 negatives	NUM: NEC-C5	NUM: SUFF-C5	largest

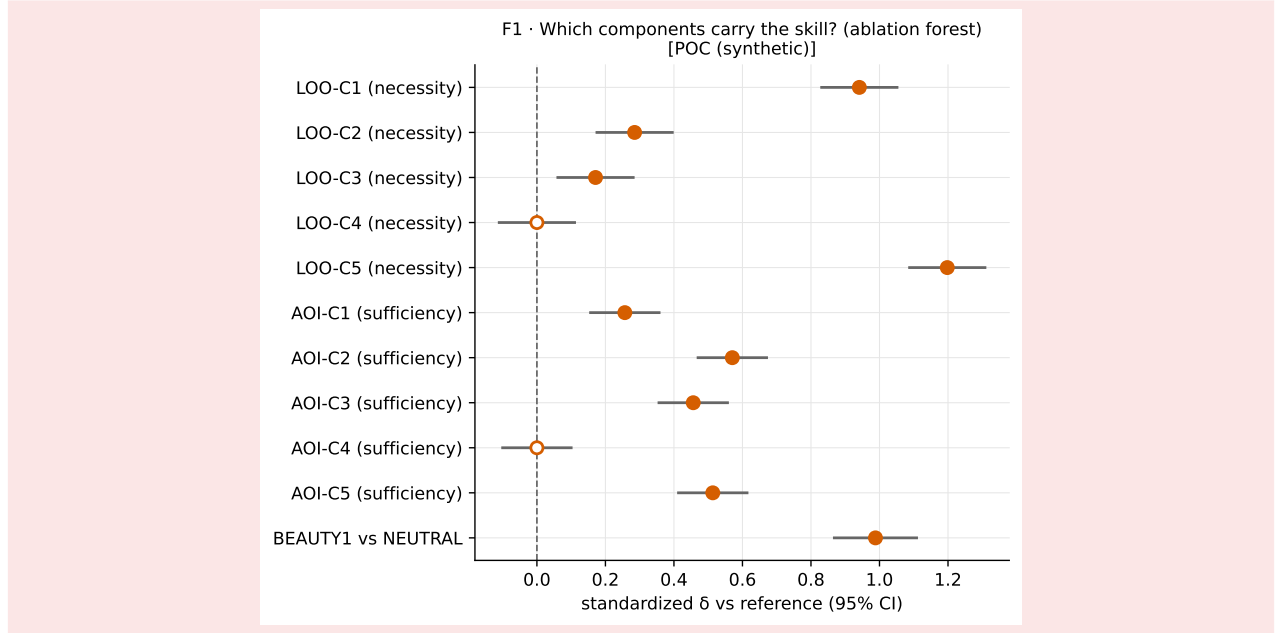


Figure 3: **[F1 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 5).* Which components carry the skill? Standardized δ vs. reference for each LOO (necessity), AOI (sufficiency), and BEAUTY1 cell, with Holm CIs; filled markers survive Holm. The twin-panel POC/BT layout is final; the effect sizes are planted (C5, C1 largest; C4 near-null by construction).

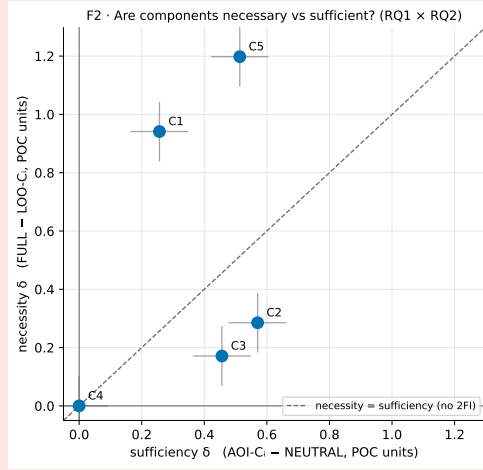


Figure 4: **[F2 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 5).* Necessity $\times$ sufficiency. Points above the $y=x$ line are more necessary than sufficient—positive interaction load (Eq. 1). Planted coordinates; the real figure places the five components by their measured effects.

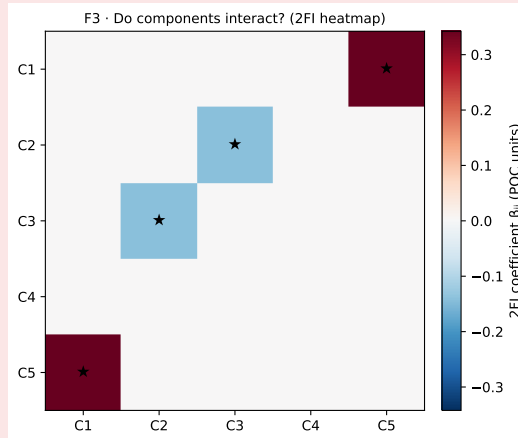


Figure 5: **[F3 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 5).* Two-factor interaction coefficients (factorial MixedLM), BH-significant cells starred. Planted $C1 \times C5$ super-additivity and $C2 \times C3$ sub-additivity illustrate the encoding; real signs and magnitudes replace them.

HF steered output. We reuse the Stage-1 FULL and NEUTRAL dev generations (200 matched pairs) as the extraction corpus, capturing the mean residual over response tokens by teacher-forced forward pass.

Why the residual stream, and why diff-in-means. Qwen2.5-Coder is a pre-norm decoder: each block reads a LayerNorm of the stream and writes back additively, so the residual stream is a linear channel where a concept can be *read* by projection and *written* by addition. The clean-design direction at layer ℓ is the difference of response-mean residuals,

$$v_\ell = \bar{h}_\ell^{(\text{FULL})} - \bar{h}_\ell^{(\text{NEUTRAL})}, \quad \hat{v}_\ell = v_\ell / \|v_\ell\|. \quad (8)$$

The contrast against the *length-matched* neutral is what makes v_ℓ “design content” rather than “instruction-following mode.” We steer with this un-whitened vector rather than a probe direction on purpose: for equal-covariance Gaussians the optimal *discriminant* is $w^* = \Sigma^{-1}(\mu_1 - \mu_0)$, ideal for *reading* but pathological for *writing*—it amplifies low-variance nuisance directions and pushes activations off-distribution—whereas diff-in-means is the empirical mean shift that lands on the data manifold and cancels nuisance directions common to both sides (Appendix C). We cross-check $\hat{v}_\ell$ against the PCA-of-paired-differences top component and the probe weight, reporting their cosines.

Step 1 — stability and localization (RQ5). Extraction adequacy is a running-mean cosine plateau: as n grows to 200 we require $\Delta \cos < 0.01$ over the last 20% (observed `NUM:STAB-DCOS` at $n = \text{NUM:STAB-PLATN}$; Figure 6). Localization triangulates three signals per layer: a linear probe’s AUC, its control-task *selectivity* $\text{AUC}^{\text{real}} - \text{AUC}^{\text{control}}$ (a probe can memorize; selectivity guards against it), and denoising activation-patching %-recovered on a design-token logit-diff proxy (denoising is robust to self-repair). The band is the contiguous layers with $\text{AUC} \geq 0.80$, $\text{selectivity} \geq 0.15$, and $\text{patch} \geq 25\%$ (Figure 7).

OUTCOME RQ5 localized

The three signals concur on a contiguous mid-band, `NUM:LOC-BAND`: probe AUC peaks at `NUM:LOC-AUCPEAK` (layer `NUM:LOC-PEAKLAYER`) with selectivity `NUM:LOC-SELPEAK` and patching recovering `NUM:LOC-PATCHPEAK` of the clean-corrupted gap. Decodability, selectivity, and causal patching agreeing at the same depth is the clean case; these layers become the Stage-2 focus band. Decodability alone would be weak evidence—a probe can read a feature the model does not use—so the patching agreement is what licenses moving from correlation to a causal candidate.

OUTCOME RQ5 diffuse

No band meets all three thresholds simultaneously—for instance, probe AUC is high across many layers but selectivity or patching never clears its bar, indicating the signal is decodable but not localized to a causal site. We widen to the single peak layer, flag the representation as diffuse, and pre-commit to multi-layer injection in the steering step. A diffuse locate is not yet a null: it changes the steering *mechanism* (where to inject), and the causal test in Step 3 remains decisive.

Step 2 — steer, and freeze on dev (RQ6 operating point). Steering adds the unit direction at all generated positions,

$$h_\ell \leftarrow h_\ell + \alpha \hat{v}_\ell, \quad \alpha = \rho \cdot \overline{\|h_\ell\|}, \quad (9)$$

parameterized by the norm-relative ρ so the coefficient transfers across layers (the ambient residual norm grows with depth; a fixed ρ injects a fixed fraction of it). Behavior is *non-monotone* in ρ —too much steering leaves the manifold and coherence collapses—so we sweep ρ under a coherence guardrail rather than fixing a large value. On a frozen 512-token eval text we cap the mean per-token divergence of the steered from the unsteered next-token distribution, $D_{\text{KL}}(p^{\text{steered}} \| p^{\text{unsteered}}) \leq 0.30$ nats, and require steered render-success ≥ 0.90 . The operating point is selected on *dev only*:

$$(\ell^*, \rho^*, \text{variant}^*) = \arg \max_{\text{dev}} \text{POC-gain} \quad \text{s.t.} \quad \text{mean-KL} \leq 0.30 \text{ and render-success} \geq 0.90,$$

ties broken toward lower KL. The dev sweep is Figure 8 (POC-gain over layer $\times\rho$, guardrail-violating cells hatched) and the dose-response is Figure 9. The frozen point is $\ell^* = \text{NUM:STEER-LSTAR}$, $\rho^* = \text{NUM:STEER-RHOSTAR}$, variant `NUM:STEER-VARIANT`, with dev POC-gain `NUM:STEER-DEVGAIN` at mean KL `NUM:STEER-KL` and render-success `NUM:STEER-RENDER`. It is written to a config file, hashed, and git-tagged *before* any held-out generation; the held-out set is touched exactly once. If no single-layer config clears the guardrail-constrained threshold, we escalate to multi-layer injection or a 2-4D subspace (pre-stated fallbacks) and re-select.

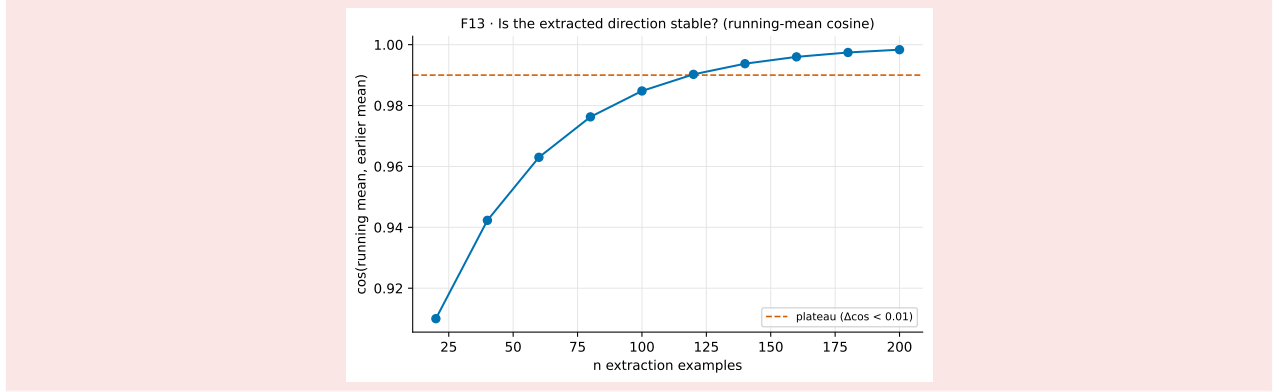


Figure 6: **[F13 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 6).* Extraction stability: cosine of the running-mean direction vs. number of examples, with the $\Delta \cos < 0.01$ plateau line. Planted convergence; the real curve certifies the 200-pair corpus (or triggers the seed-extension contingency).

Step 3 — the causal bar (RQ6), stated as interventions. Steering and ablation are do-operations on an activation node, severing its normal dependence on upstream computation. The two arms are

$$\text{ sufficiency: } \mathbb{E}[Q \mid \text{do}(h_{\ell^*} + = \alpha^* \hat{v}_{\ell^*}), \text{NOSYS}] > \mathbb{E}[Q \mid \text{NOSYS}], \quad (10)$$

$$\text{ necessity (flip): } \mathbb{E}[Q \mid \text{do}(h \leftarrow Ph), \text{FULL}] < \mathbb{E}[Q \mid \text{FULL}], \quad P = I - \hat{v}\hat{v}^\top, \quad (11)$$

with Q the POC or a preference utility. Sufficiency uses the NOSYS base—*no design prompt*—so the effect is the direction’s, not an instruction’s; that is what makes it interventional rather than observational. Necessity projects the direction out at every layer and position while the skill is present (weight-orthogonalization, so no inference hook is needed), asking whether the skill stops working. The projector P is a genuine orthogonal projector (idempotent and symmetric; Appendix C). Held-out arms are unsteered-NOSYS, steered-NOSYS, a norm-matched random control ($k=5$ draws re-normalized to $\|v_{\ell^*}\|$), and an HF-generated FULL reference; we report both oracle signals and the reproduction fraction $\% \text{skill-reproduced} = (\text{POC}_{\text{steered}} - \text{POC}_{\text{unsteered}}) / (\text{POC}_{\text{FULL}} - \text{POC}_{\text{unsteered}}) \times 100\%$.

The success criterion (verbatim). The direction is a *causally sufficient design direction* iff, on held-out prompts (including 10 unseen-type):

steered-NOSYS beats unsteered-NOSYS on ≥ 1 objective family after Holm AND the reliability-gated preference delta (steered $\succ$ unsteered) has BT-utility cluster-bootstrap CI > 0 ; AND the norm-matched random control ($k=5$ draws) does not satisfy both (specificity); AND the flip test attenuates the skill’s gain (necessity). If addition succeeds but the flip test does not, report “sufficient, not demonstrably necessary.”

This four-conjunct standard is why a *causal* headline is defensible: under a true null the joint false-positive rate is below any single test’s α , and the four checks probe independent failure modes—objective vs. preference vs. specificity vs. necessity (Appendix C). The specificity conjunct formalizes non-identifiability: if a random direction of equal norm also lifts Q , the effect is a *norm* effect, not a *direction* effect. We additionally bound side-effects: steering must not drop non-UI algorithmic pass-rate by $> 10\%$ absolute nor push general-text mean KL > 0.30 nats.

7 Stage 2 results: does the direction steer?

This is the pivotal section. With the operating point frozen on dev and git-tagged, we touch the held-out set once and evaluate the four-conjunct criterion. The held-out arms—unsteered-NOSYS, steered-NOSYS, norm-matched random ($k=5$), and an HF FULL reference—are compared on both oracle signals in Figure 10; steering

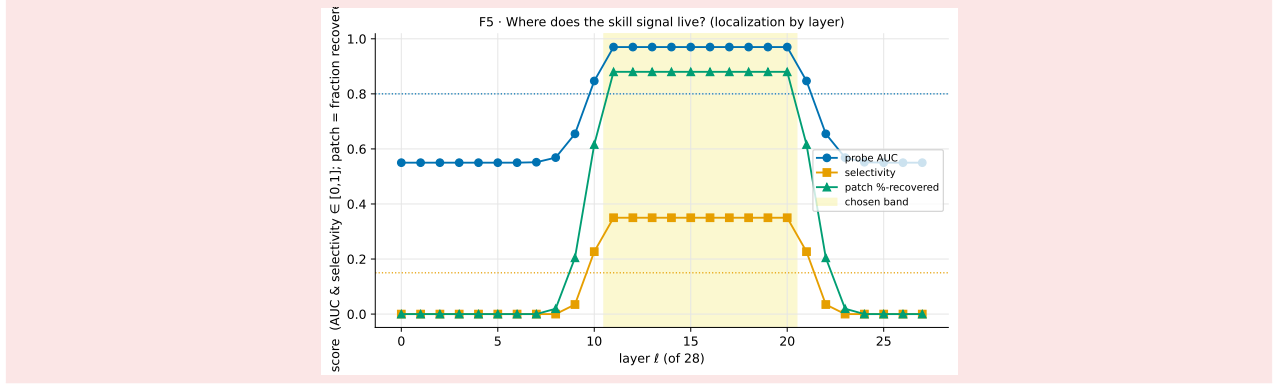


Figure 7: **[F5 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 6).* Where the skill signal lives: probe AUC, control-task selectivity, and patch %-recovered by layer, with the chosen band shaded and the AUC/selectivity thresholds drawn. Planted mid-band plateau; the real curves set the Stage-2 focus layers (RQ5).

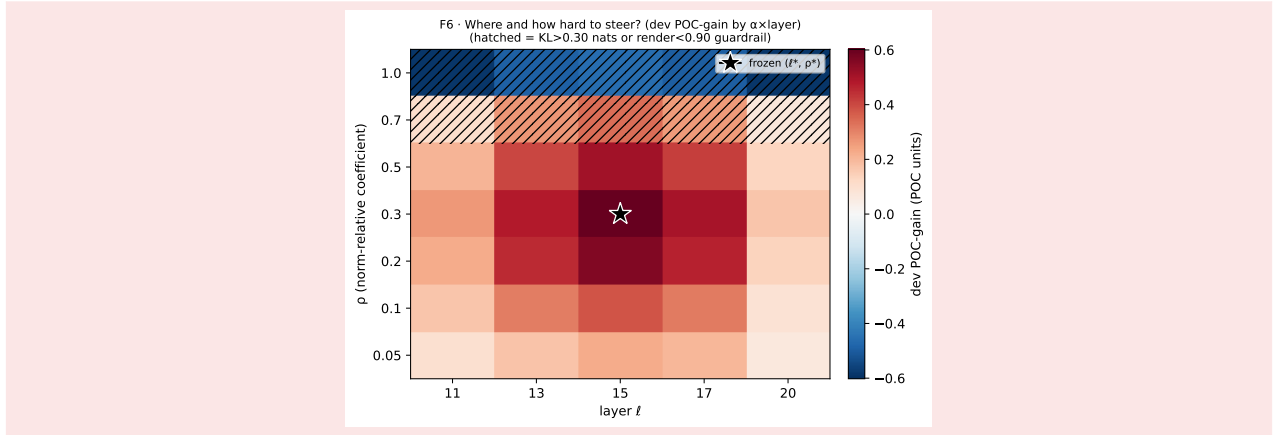


Figure 8: **[F6 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 7).* Dev sweep: POC-gain over layer $\times \rho$; guardrail-violating cells ($KL > 0.30$ or $render < 0.90$) hatched; the frozen (ℓ^*, ρ^*) starred. The operating point is selected here, on dev, then git-tagged before held-out. Planted optimum.

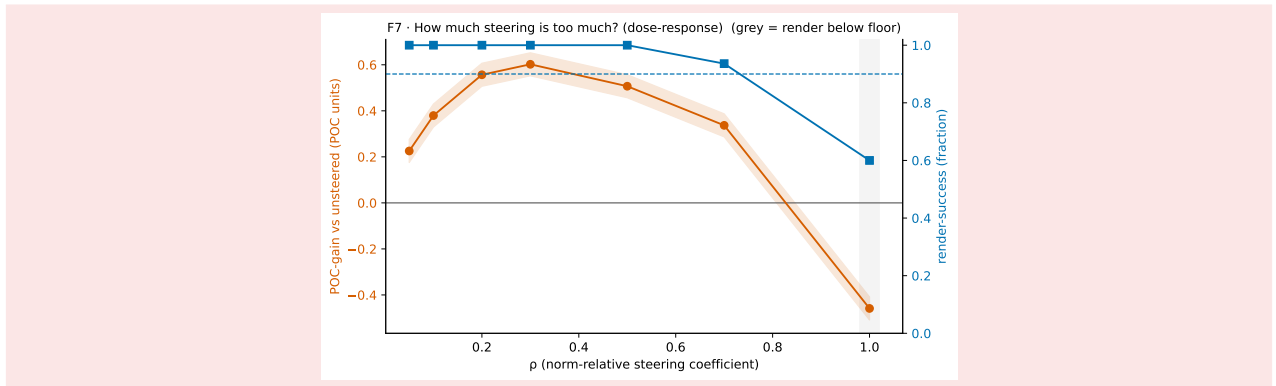


Figure 9: **[F7 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 7).* Dose-response at ℓ^* : POC-gain (with CI band) and render-success vs. ρ , failure region shaded. The non-monotone gain then coherence collapse is the expected shape; planted here, measured on Colab.

reproduces `NUM:STEER-REPRO` of the skill’s POC gain. The result resolves into one of three preregistered branches, written out in full below. Exactly one is kept on real data.

7.1 Branch A — a causally sufficient (and/or necessary) design direction exists

OUTCOME A: causally sufficient direction

Steering the diff-in-means direction with *no design prompt* raises held-out UI quality on both oracle signals: steered-nosys beats unsteered-nosys by `NUM:STEER-DELTA` on the POC (significant on `NUM:STEER-NFAM` of six objective families after Holm) and is preferred with BT utility `NUM:STEER-BT` (cluster-bootstrap CI > 0). Crucially, the norm-matched random control does not clear both bars (`NUM:STEER-RAND` POC, CI bracketing zero), so the effect is a *direction* effect, not a *norm* effect—a random push of equal magnitude does not make pages better. The direction reproduces `NUM:STEER-REPRO` of the full skill’s held-out gain (Figure 10), and it does so on prompts and *task types the extraction corpus never contained*. To our knowledge this is the first demonstration of a causal, steerable direction for the visual/aesthetic quality of generated front-end code: a few sentences of design guidance and a single mid-network vector turn out to move the same underlying quantity, and that quantity can be actuated with the prose removed.

Coherence and specificity. The gain is not bought with gibberish: at ρ^* the mean next-token KL from the unsteered model is `NUM:STEER-KL` nats (under the 0.30 guardrail) and steered render-success is `NUM:STEER-RENDER`. Steering leaves general ability intact—non-UI algorithmic pass-rate drops by only `NUM:SPEC-PASSDROP` (within the 10% tolerance) and general-text KL is `NUM:SPEC-KL`—so the direction is specific to design, not a blunt instrument that degrades everything.

Necessity (the flip test). Projecting $\hat{v}$ out at every layer and position while the skill is present attenuates the skill’s own gain by `NUM:FLIP-ATTEN` (Eq. 11), which promotes the claim from sufficiency to necessity.

OUTCOME A.1: sufficient and necessary

The flip test attenuates the skill’s gain by `NUM:FLIP-ATTEN` with CI > 0: removing the single direction from the residual stream substantially disables the skill even when its 391 tokens are still in context. The direction is therefore both sufficient (adding it with no prompt helps) and necessary (removing it stops the prompt from helping)—the strong two-arm result, mirroring the refusal-direction template but for a graded, multi-component aesthetic behavior. This is the paper’s strongest mechanistic claim: the skill’s effect is, to a large extent, *routed through this one direction*.

OUTCOME A.2: sufficient, not demonstrably necessary

Addition succeeds but the flip test does not attenuate the skill’s gain (`NUM:FLIP-ATTEN`, CI includes zero): the direction is a sufficient carrier but not the unique one. The skill routes its effect through this direction *and* others—consistent with a concept cone rather than a single axis. We report “sufficient, not demonstrably necessary” and do not overclaim uniqueness; the honest framing is “a causally sufficient design direction,” never “*the* design direction.”

Failure surface and generalization (RQ7). A positive mean can hide heavy-tailed per-prompt behavior, so we report the distribution. The overall anti-steerable fraction (prompts whose steered POC falls below unsteered) is `NUM:FAIL-ANTI-OVERALL`; on unseen types it is `NUM:FAIL-ANTI-UNSEEN`, and the mean unseen-type steering delta is `NUM:FAIL-UNSEEN-D` (Figure 11). The reproduction fraction on unseen types is `NUM:FAIL-REPRO-UNSEEN`.

OUTCOME RQ7 robust

The anti-steerable fraction stays below 50% across all task types, seen and unseen, and the unseen-type delta is positive and gated: the direction transfers cleanly out-of-distribution. Steering a direction extracted only from marketing- and content-oriented dev pages nonetheless improves control-dense settings panels and admin tables, evidence that it encodes transferable design *content* rather than a task-idiom artifact.

OUTCOME RQ7 partial

Steering helps on average and generalizes to most unseen types, but a named regime is brittle: `NUM:FAIL-WORST-TYPE` crosses the 50% anti-steerable line (`NUM:FAIL-WORST-ANTI`), so steering makes those pages worse as often as better (Figure 11). This is the honest failure map—the direction is not a blanket win, and its data-dense failure mode is exactly where the aesthetic prior (generous whitespace, one signature element) conflicts with the task’s information density.

OUTCOME RQ7 brittle

The unseen-type effect collapses (anti-steerable $\geq 50\%$ overall on OOD): the direction is dev-specific and does

not transport to genuinely new layouts. We scope the causal claim to the in-distribution regime and report the OOD boundary precisely; the direction is real but local, and its extraction corpus bounds where it works.

Component↔direction correspondence (RQ8). Finally we test whether the component structure the ablation found survives into the representation. Component sub-vectors $v_{C_i} = \bar{h}_{\ell^*}^{(\text{AOI-}C_i)} - \bar{h}_{\ell^*}^{(\text{NEUTRAL})}$ have mean off-diagonal cosine `NUM:CORR-COSMEAN` (max `NUM:CORR-COSMAX`), and `NUM:CORR-NMATCH` of five steer their own metric family (Figure 12). Cosine can show sub-vectors are linearly distinguishable but not that each *causally* steers its family, so the rule pairs near-orthogonality with a steering-signature match.

OUTCOME RQ8 found

The sub-vectors are near-orthogonal (mean $|\cos| = \text{NUM:CORR-COSMEAN} < 0.3$) and $\geq 3/5$ steer their own metric family—C1 moves color, C2 moves layout, and so on. The design skill has a *component basis* inside the model, and the black-box ablation signatures correspond to white-box steering directions: the paper’s strongest bridge result, an ablation↔activation correspondence predicted by the geometry of linear representations. It means “clean design” is not one entangled thing but a small set of separable, individually actuatable sub-concepts.

OUTCOME RQ8 partial

One or two sub-vectors steer their own family while the rest are entangled (`NUM:CORR-NMATCH`/5 match): the representation has partial component structure. Some components (likely color and layout, the most necessary) have distinct directions; others share a subspace. The bridge holds in part—the most causal components are the most separable—but “clean design” is not a fully diagonal basis.

OUTCOME RQ8 absent

The sub-vectors are not near-orthogonal, or fewer than three steer their own family: there is no component basis. “Clean design” is one entangled direction inside the model even though the prompt-level skill decomposes into five components. This is still informative and slightly surprising: the ablation’s component structure is a property of the *instruction*, not of the model’s internal representation—the model compresses five instructions into one axis of variation.

7.2 Branch B — the honest null

OUTCOME B: not linearly steerable

No single linear direction steers held-out UI quality under the coherence guardrail. Steered-NOSYS does not beat unsteered-NOSYS on any objective family after Holm (`NUM:STEER-DELTA` POC, CI `NUM:STEER-DELTA-CI` bracketing zero) and the preference delta’s BT CI includes zero (Figure 10). This is a *null with teeth*, not an absence of evidence: the power gate certifies the confidence intervals are tight enough to exclude effects the study was designed to detect, and the extraction direction *is* decodable (RQ5 probes separate FULL from NEUTRAL at high AUC)—so we have a direction that *correlates* with design but does not *cause* it when injected. A correlational direction that does not steer is a hypothesis, not a finding; we report it as such.

The evidential value is precise. The null *rules out* the strongest form of the linear-representation account for this behavior in this model: that a single mid-network direction, added with no prompt, suffices to reproduce the skill’s effect. It is consistent with two readings we cannot distinguish here and state honestly: either the skill’s effect is *distributed* across many directions and/or layers (a single injection is too blunt—the pre-stated multi-layer and subspace fallbacks were tried and also failed), or the behavior is genuinely *non-linear* in the residual stream (diff-in-means, a linear probe of a nonlinear model, cannot capture it). Either way, the practical implication is that the design skill’s mechanism is not a simple steerable feature: the 391 tokens do more than trigger one latent switch. For the interpretability literature, it is a negative datapoint on how far the single-direction picture—so clean for refusal—extends to graded, compositional, generation-spanning aesthetic behaviors. We publish it with the same rigor as a positive result, because a forced or buried null would corrupt exactly the oracle this project is built to defend.

7.3 Branch C — partial outcomes

Between a clean causal direction and a clean null lie four partial results, each shipped with its numbers and each carrying a distinct, bounded claim.

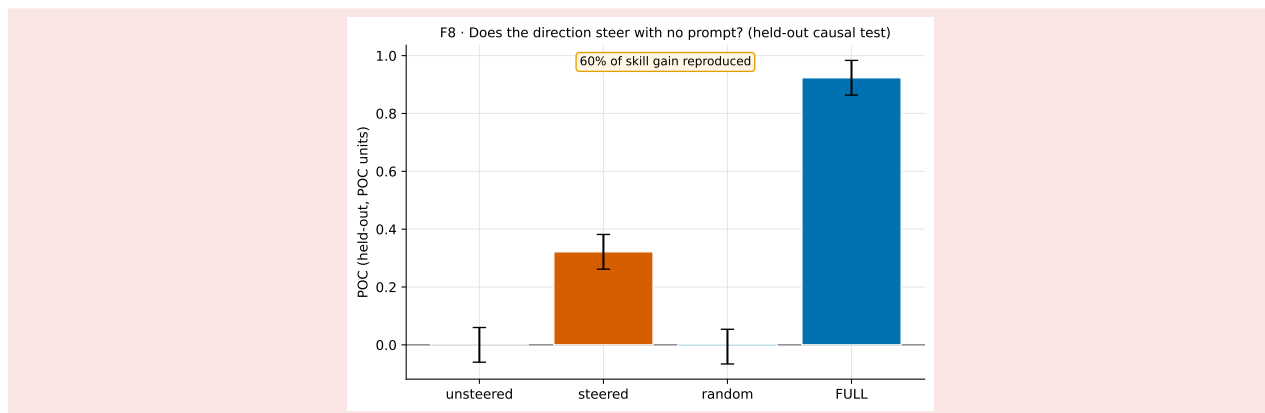


Figure 10: **[F8 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 8).* The held-out causal test. POC by arm—unsteered, steered (no design prompt), norm-matched random ($k=5$), and FULL reference—with the %-skill-reproduced annotation. Success requires steered > unsteered while random $\approx$ unsteered. Planted arms illustrate the layout; the real bars decide RQ6.

OUTCOME C.1: steers the objective signal only

Steered-NOSYS beats unsteered on ≥ 1 objective family after Holm (`NUM:STEER-DELTA`), but the preference delta’s BT CI includes zero. The direction moves the *measurable* correlates of design—contrast, alignment, slop tells—without moving *human-judged* quality. Given that objective metrics cap at $\sim$ half the variance in appeal, this is a meaningful boundary: we have steered what the metrics see but not what a designer prefers, and by the null rule the aesthetic claim is inadmissible. We report a “metric-level” steering effect and explicitly decline the quality claim.

OUTCOME C.2: steers, but only by breaking coherence

A configuration raises POC only above the KL guardrail (`NUM:STEER-KL` nats > 0.30) or below the render floor: the direction “improves” pages by pushing the model off-distribution, and the gain rides on artifacts rather than design. Under the preregistered guardrail this does not count as success. We report the direction as *coherence-breaking* and show the dose-response (Figure 9) where the apparent gain and the KL blow-up coincide—an honest account of a non-result that a laxer guardrail would have mis-sold as a finding.

OUTCOME C.3: anti-steerable majority

The mean steering effect is near zero but hides a heavy-tailed distribution: `NUM:FAIL-ANTI-OVERALL` of prompts are anti-steerable, so steering helps half and harms half (Figure 11). The “average effect” is an artifact of cancellation, and reporting only the mean would be actively misleading—the reason the protocol reports the full per-prompt distribution. The direction is not a reliable design lever; whatever it encodes is prompt-contingent in a way a single global vector cannot resolve.

OUTCOME C.4: sufficient but not necessary

Addition succeeds on both signals (Branch A holds) but the flip test does not attenuate the skill’s gain (`NUM:FLIP-ATTEN`, CI includes zero): a steerable direction exists, yet the skill does not *depend* on it—the model has other routes to the same behavior. This is the concept-cone outcome; we claim a sufficient direction and an honest account of non-uniqueness, and we present the cosine/cone structure (Figure 12) rather than a single-axis story.

8 Discussion

The bridge: does the black-box explanation match the white-box one? The study’s distinctive move is to ask one question two ways—which *instruction* matters (Stage 1) and which *direction* matters (Stage 2)—on the same model, and then to check whether the answers agree. That check is the correspondence between the components RQ1 finds most necessary and the sub-directions RQ8 finds separable.

OUTCOME bridge holds

If RQ1 ranks C5 (negatives) and C1 (color) most necessary *and* RQ8 finds those the sub-vectors that most cleanly steer their own metric families, the two explanations converge: the parts of the prose that carry the causal

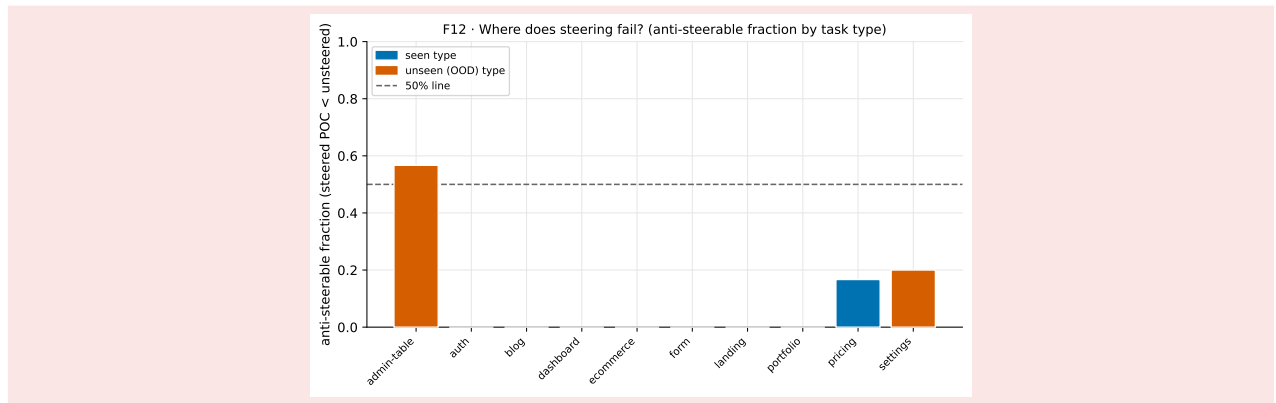


Figure 11: **[F12 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 8).* Anti-steerable fraction by task type (seen vs. unseen/OOD), with the 50% brittleness line. A positive mean can coexist with a brittle unseen type; the distribution, not the mean, is the RQ7 claim. Planted; real data replaces the bars.

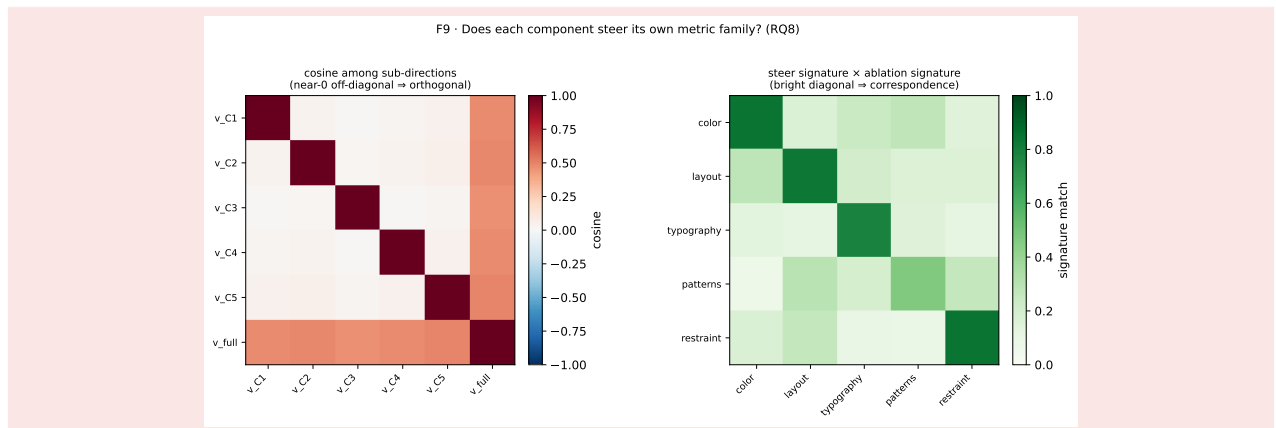


Figure 12: **[F9 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 9).* Component $\leftrightarrow$ direction correspondence (RQ8). Left: cosine matrix among the five component sub-vectors and v_{full} (near-zero off-diagonal $\Rightarrow$ orthogonal). Right: steer-signature $\times$ ablation-signature match (a bright diagonal $\Rightarrow$ each component steers its own family). Planted diagonal; real data decides found/partial/absent.

weight are the parts with the crispest internal directions. That convergence is the paper’s most satisfying result—a prompt-level account and a representation-level account of the *same* phenomenon, mutually corroborating—and it suggests skill authors are, in effect, writing instructions that map onto the model’s existing axes of variation.

OUTCOME bridge fails

If the most necessary components are *not* the most separable directions (RQ8 absent or mismatched), the two levels tell different stories: the instruction’s compositional structure is not the model’s. This is equally interesting—it says the componentization that helps a human author a skill is not how the model represents design—and it cautions against reading prompt structure as mechanism.

What “make it beautiful” tells us about instruction compression. RQ4 turns a cheap one-liner into a probe of how much of a 391-token skill is *knowledge* versus *permission*. If a five-word nudge recovers most of the effect (RQ4 nudge), the model already contains the capability and the skill largely licenses its use—which is precisely the premise that makes Stage 2 plausible: a capability the model already has is one a direction can actuate. If instead the structured content is load-bearing (RQ4 content), the skill transfers specific knowledge—named palettes, type scales, prohibitions—and we should expect the extracted direction to be correspondingly richer than a single “try harder” axis. Either way, RQ4 and Stage 2 are two readings of the same question: is the design skill flipping a switch the model already has, or installing something new?

Mechanistic implications. A positive Stage 2 result would extend the single-direction picture—clean for refusal [Arditi et al., 2024]—to a graded, compositional, generation-spanning *aesthetic* behavior, which is a substantially harder case than a binary safety behavior: the target is downstream of thousands of tokens of code and has no single “answer token.” A null would mark a boundary of that picture, and the honest reading is that not every behavior a prompt induces is a steerable feature. The correspondence result (RQ8), whichever way it falls, speaks to the geometry of compositional concepts [Park et al., 2024a]: whether a human’s five-way decomposition of “design” lines up with the model’s internal near-orthogonal directions.

Relation to the steering-reliability literature. We inherit its central warning: mean effects hide heavy tails, and up to half of inputs can be anti-steerable [Tan et al., 2024, Queiroz Da Silva et al., 2025]. Our protocol answers it directly—per-prompt distributions, an anti-steerable fraction, a class-separation pre-screen, an OOD held-out over unseen *types*, and a coherence guardrail on a non-monotone dose-response [Taimeskhanov et al., 2026]. The four-conjunct success criterion, in particular, is designed so that a strong average cannot by itself buy a causal claim: specificity and necessity must also hold. This is the methodological transfer we hope outlasts the particular numbers—a template for claiming a *causal* steering result responsibly when the underlying phenomenon is known to be unreliable.

Practical guidance for skill design. Read together, the Stage-1 outcomes yield actionable advice for authors of design skills: invest tokens where necessity concentrates (the RQ1 ranking), prune any component whose leave-one-out is null or negative, and exploit the interactions RQ3 surfaces—if $C1 \times C5$ is super-additive, a palette instruction and a “no purple gradients” prohibition should ship together, not separately. If sufficiency is distributed (RQ2), resist the urge to shorten the skill to its single largest component; if it is concentrated, the skill can be compressed. These are empirical recommendations, contingent on which branch obtains, and each is traceable to a preregistered contrast.

9 Limitations and threats to validity

We state the bounds of the study up front, and pair each internal-validity threat with the control that neutralizes it.

One model, one family. Every result is about Qwen2.5-Coder-7B-Instruct. The two-stage design *requires* a single open-weights model so that the ablation and the interpretability describe the same system; the price is that generalization to other models and scales is untested. This is a deliberate scope choice, not an oversight, and the recorded fallback model would restart both stages rather than mix them.

One skill instantiation. We ablate one five-component skill with one content-orthogonal decomposition (C1–C4 prescriptive, C5 all-negatives). The componentization is one defensible cut among several—a different author might split typography from type-scale, or fold patterns into layout—and the necessity/sufficiency verdicts are conditional on this cut. We fix and document the decomposition and its token balance, and note that RQ8 tests whether *this* cut corresponds to internal structure; a different cut could correspond differently.

The judge is not a human panel. The preference signal is a VLM judge validated on a 100-pair human subset and gated on chance-corrected agreement, but it is not a full human study. Its known biases (length, verbosity, position) are controlled by both-orders judging and style-controlled Bradley–Terry, and the objective half is the unfoolable anchor—but the preference channel’s ceiling is the judge’s fidelity, which the reliability gate bounds rather than eliminates. Where the two $\hat{\beta}$ (controlled vs. uncontrolled) disagree in sign, we report the contrast as style-confounded rather than resolved.

Objective metrics cap at half the variance. Interface-aesthetics metrics explain $\lesssim$ half the variance in human appeal [Miniukovich and De Angeli, 2015, Reinecke et al., 2013]. We therefore never let the objective half stand alone for an aesthetic claim; it is necessary (deterministic, unfoolable) but not sufficient, which is the whole reason for the second signal. The PSI is validated against human “AI-slop” ratings before use, and demoted to a 6-term POC if it fails.

Diff-in-means is a linear probe of a nonlinear model. A null steering result bounds *linear* steerability at the chosen sites, not the existence of the concept: the behavior could be non-linear [Engels et al., 2024] or distributed across a subspace or cone [Wollschläger et al., 2025]. We say so explicitly, try the multi-layer and subspace fallbacks before declaring a null, and frame any positive result as “*a* sufficient direction,” never “the” direction—non-identifiability is assumed, and the specificity control ($k=5$ random draws) is what separates a direction effect from a norm effect.

The interpretability threats, and their controls. Table 4 lists the Stage-2 threats and the control that neutralizes each; these are built into the design, not applied afterward.

Table 4: Stage-2 internal-validity threats and controls (Table T9, abridged).

threat	control
Norm confound (any large push jogs quality)	norm-matched random vector, $k=5$ draws; must not reproduce the gain
Non-identifiability / concept cones	claim “a” not “the” direction; report cosine/cone structure (RQ8)
Self-repair inflating patching necessity	<i>denoising</i> (not noising) activation patching
Interpretability illusion (dormant pathway)	triangulate probing + patching + steering (RQ5 rule)
Spurious extraction cues	held-out incl. unseen task <i>types</i> ; class-separation pre-screen
Selection leakage (tuning on test)	freeze (ℓ^* , ρ^* , variant) on dev, git-tag before held-out, touch once
Engine confound (vLLM vs. HF)	within-HF comparisons only; re-generate baselines in HF

Statistical scope. Cluster-bootstrap CIs are calibrated to the *number of prompts* (40 dev, 18 held-out), not the number of generations, so their tails are limited at 40 clusters; we report parametric MixedLM CIs as a robustness pair and do not claim generalization beyond the 58-brief task population. Post-hoc power is used only as an admissibility screen, never to reinterpret an under-powered delta as a null.

Compute scale. The GPU budget is modest (a few L4- and A100-hours; Section 10), which fixes the corpus and seed counts. Larger corpora or more seeds would tighten every CI; the design allocates its five headline seeds and 40 dev prompts to the confirmatory contrasts and accepts lighter power on the exploratory scans, controlled by BH.

10 Reproducibility and ethics

Preregistration and freezes. All confirmatory decisions—the cell table, the 58-brief split, the POC formula and its two conditionals, the 12-test confirmatory family, the reliability and power gates, the Stage-2 four-conjunct success criterion, and the null rule—are frozen in a preregistration (Appendix F) git-tagged `prereg-v1` before the first GPU run. The Stage-2 operating point (ℓ^* , ρ^* , variant*) is selected on dev, written to a hashed config, and git-tagged `steer-frozen` before the held-out set is touched; held-out is evaluated exactly once. Any post-freeze change is a dated amendment; a confirmatory change forks a new tag and is reported as a deviation.

Determinism and pinning. An engine split is enforced: vLLM for Stage-1 bulk generation, HF for all activation and steering work, never cross-compared; engine, version, sampling, and config hashes are stamped on every results row. Sampling is fixed (Section 3). Rendering is deterministic: pinned Chromium at 1440×900 (plus a 390×844 mobile subset for the responsiveness slice), device-scale-factor 1, animations disabled, fonts-ready awaited, network blocked; a metric artifact re-run under an unchanged config reproduces identical hashes. All pinned versions are recorded in an environment lock.

Artifacts and machinery. Reusable machinery lives in a tested Python package (metrics, rendering, judging, agreement, Bradley–Terry, mixed models, power, hooks, steering, figures, manifest); notebooks carry narrative and orchestration and import from it. The CPU-safe half—fixture rendering, the full objective-metric stack, the judge in dry-run/mock mode, the statistics on synthetic data, and every unit test—runs and passes in the development environment; the GPU-dependent half is written, mock-tested, import-guarded, and left unexecuted until the Colab sessions in the runbook. Every figure in this paper is produced by the same figure module from a results table—no hand-editing—and the preview figures here are that module run on synthetic data with a planted ground truth (unit-tested to recover the plant). An artifact manifest records every HTML, screenshot, DOM-JSON, activation shard, and results file with a SHA-256, its producing config hash, engine, and commit.

Cost. The full study is deliberately inexpensive (Table 5), which we regard as a feature: a two-stage causal study of a design skill fits in a graduate compute budget.

Table 5: Compute and API budget (Table T7). These are planned budgets, not results.

item	quantity	cost
Stage-1 generation (vLLM, L4)	~6 L4-hours	≈243 Colab units, \$50–65
Stage-2 capture/sweep/verify/stretch (A100)	~14 A100-hours	
Judge — Gemini 2.5 Flash primary	~4,420 both-order calls	≈\$7
Judge — GPT-4o audit (15%)	~664 calls	≈\$7
Storage (HTML/PNG/DOM-JSON/activations)	~6,900 units + means	≈4–5 GB
Total		≈\$65–80

Ethics. The study generates web-UI code and evaluates its visual quality; it involves no personal data and no human subjects beyond the authors’ own labeling of a 100-pair validation subset (no demographic data, no external participants). The subject model is used under its license; API judges are used only as evaluators. The one dual-use-adjacent technique is weight-orthogonalization, used here solely as the necessity flip test on an *aesthetic* direction, with no safety-relevant behavior involved. We see no additional ethical concerns.

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A The design skill, fillers, and padding pool (verbatim)

The frozen independent variable. Component boundary tags `[[Cn]]` are documentation only and are stripped before the text enters the system prompt; canonical order is $C1 \rightarrow C2 \rightarrow C3 \rightarrow C4 \rightarrow C5$. $C1$ – $C4$ are purely pre-scriptive; $C5$ consolidates all negative constraints.

C1 — color / palette.

Choose a deliberate palette of four to six specific colors — one dominant hue, one or two supporting tones, and one sharp accent — each fixed as an exact hex value and reused consistently throughout. Build depth with atmospheric, layered backgrounds such as subtle tints, soft shading, or fine texture rather than flat fills, and ensure every text-on-background pairing clears WCAG AA contrast.

C2 — layout / spacing / hierarchy.

Open with the most characteristic element of the subject so the page announces what it is at a glance. Establish a clear visual hierarchy on a consistent spacing scale, aligning elements to a small set of shared edges so the composition reads as engineered. Favor intentional, sometimes asymmetric arrangements over uniform grids, and give the layout generous, purposeful white space so it can breathe.

C3 — typography.

Pair a characterful display typeface, used sparingly for headings, with a clean, highly readable body typeface and a compact utility face for small labels and data. Set a clear modular type scale with a few deliberate sizes and weights, and let the typography carry real personality — expressive at large sizes, quietly legible at small ones — instead of neutrally delivering text.

C4 — component patterns.

Design one well-chosen signature element — an unexpected section, a distinctive card treatment, or a considered navigation — that anchors the page's identity. Add at most one orchestrated motion moment, such as a single page-load reveal or a refined hover state, and make it feel deliberate. Vary corner radii, borders, and shadows with intent so that depth and emphasis track the importance of the content.

C5 — negative constraints (all negatives).

Avoid AI slop defaults: purple or indigo gradients; Inter and Roboto typefaces; fully centered hero layouts built around one big headline and stat; three identical icon cards in a row; uniform rounded corners on everything; and scattered or excessive animation. Never default to generic template palettes or cream with terracotta serif and near black with acid green looks, unless the brief calls for them. Cut any decoration that serves no purpose.

Neutral fillers (render-inert; matched to C1–C5). A filler may touch only source-level conventions and is forbidden from mandating anything an objective metric could detect (semantic landmarks, `lang/title/meta`, ARIA, validation, event handlers), enforced by a build-time audit against a banned-topic list.

Filler-1 ($\leftrightarrow$ C1): Write the markup in a consistent source style: use lowercase for tag and attribute names, quote every attribute value with double quotes, and keep a stable attribute order within each tag. Break long lines at a consistent indent so no line grows unwieldy, and keep quoting and casing choices uniform across the whole file so the source reads cleanly.

Filler-2 ($\leftrightarrow$ C2): Indent the markup consistently with two spaces per level and keep lines to a reasonable width so the source stays legible. Group related rules together, order the CSS declarations predictably, and leave a short comment above each major part of the stylesheet describing what it styles. Avoid trailing whitespace and keep a single blank line between logical blocks.

Filler-3 ($\leftrightarrow$ C3): Name classes and identifiers with clear, lowercase, hyphen-separated words that describe purpose rather than appearance, and keep the naming scheme consistent across the file. Avoid abbreviations that are not obvious, do not reuse one name for two purposes, and prefer a short, stable vocabulary of names over inventing a new term for every element on the page.

Filler-4 ($\leftrightarrow$ C4): Organize any JavaScript into small, single-purpose functions with descriptive names, each declared before the point where it is used, and keep the logic shallow rather than deeply nested. Prefer `const` and `let` over `var`, keep variable declarations near their first use, group related functions together in the script, and leave a brief comment explaining any step whose intent is not obvious.

Filler-5 ($\leftrightarrow$ C5): Comment the code where intent is not obvious, but avoid restating what the code plainly does; a good comment explains why, not what. Keep comments short, place each one directly above the line or block it describes, and use a consistent comment style throughout. Remove commented-out fragments and leftover notes to self before finishing, so only purposeful comments remain in the final file.

Padding pool (frozen; build-time equalization). Each filler is deterministically padded (in fixed order, skipping any clause that would overshoot) or trimmed at a sentence boundary until it matches its component's Qwen token count to within ± 2 : *Pad-1a* Apply the same wrapping style to every long tag so no line stands out; *1b* Break attribute lists at a consistent width; *1c* Keep tag casing uniform throughout; *2a* Keep spacing between rule blocks even so the stylesheet reads in a steady rhythm; *2b* Order declarations the same way in every rule; *2c* Keep rule blocks tidy and short; *3a* Prefer plain descriptive words over clever coinages when a name must be introduced; *3b* Keep names short, plain, and easy to scan; *3c* Keep the vocabulary of names small; *4a* Group constants near the top of the script so their values are easy to locate; *4b* Keep the script's functions in one predictable order; *4c* Prefer flat logic over nesting; *5a* Delete stale comments rather than letting them drift out of date; *5b* Keep comment punctuation simple and consistent; *5c* Prefer one clear comment over three vague ones.

Constant output constraint (in the user turn; never ablated).

Return a single self-contained HTML file: put all CSS in one `<style>` tag and all JavaScript in one `<script>` tag, inline, with no external network requests — no external fonts, stylesheets, scripts, images, or CDNs. Use only system or generic fonts, and inline SVG or CSS for any graphics. Output only the HTML file.

Token balance. Exact Qwen2.5-Coder tokens per component: C1 75, C2 73, C3 75, C4 80, C5 88 (mean $m=78.2$, $\pm 15\%$ band [66.5, 89.9]; all within band); full skill = 391 exact tokens. Equalized fillers: F1 75, F2 74, F3 76, F4 81, F5 87 (each within 2 tokens of its component). The four control cells are FULL (C1–C5), NEUTRAL (Filler-1–5), BEAUTY1 (“Make it beautiful and well-designed.”), and NOSYS (no system message).

B Prompt corpus (frozen by id)

The 58 briefs (Table T2). Difficulty E/M/H by element count and interaction complexity. Split: **dev** (40), **seen** (held-out, seen type; 8), **unseen** (held-out, OOD type; 10). All briefs implicitly carry the constant output constraint; a whole-word leakage audit over the briefs passes (zero matches of the banned style/design word list).

id	type	brief	d	split
L01	landing	Landing page for a mobile app that tracks home water usage and flags leaks; headline, how-it-works, three feature highlights, download call.	E	dev
L02	landing	Landing page for a regional apple-cider producer selling seasonal boxes; product story, what's in a box, delivery areas, order button.	M	dev
L03	landing	Landing page for a B2B API that turns shipping documents into structured data; hero statement, integration steps, a code sample, request-access form.	H	dev
L04	landing	Launch page for an independent documentary about deep-sea mining; synopsis, trailer placeholder, screening dates, newsletter signup.	H	dev
L05	landing	Landing page for a neighborhood bicycle-repair co-op; hours, services, a pricing note, location.	E	dev
L06	landing	Landing page for a language-exchange meetup pairing learners for weekly conversation; how pairing works, supported languages, join button.	M	seen
D01	dashboard	Dashboard for a solar-panel owner; today's generation, home consumption, battery level, 7-day history chart placeholder.	M	dev
D02	dashboard	Operations dashboard for a bike-share system; station occupancy, bikes in transit, low-battery e-bikes, alerts list.	H	dev
D03	dashboard	Personal-finance dashboard; account balances, monthly spending by category, upcoming bills, savings-goal tracker.	M	dev
D04	dashboard	Dashboard for a small warehouse; inventory by aisle, orders awaiting pick, staff on shift, throughput chart placeholder.	H	dev
D05	dashboard	Dashboard for a habit-tracking app; streaks for four habits, a weekly completion grid, today's checklist.	E	dev
D06	dashboard	Fleet dashboard for a delivery company; vehicles active, deliveries completed, fuel usage, map placeholder with status pins.	M	seen
F01	form	Multi-step signup form for a coding bootcamp; personal details, program choice, payment plan, progress indicator.	E	dev
F02	form	Grant-application form for a community arts fund; applicant info, project description, budget table, file-upload fields.	M	dev
F03	form	Booking form for a pottery studio; class selection, date and time, seats, contact details.	M	dev
F04	form	Renters-insurance quote form; address, coverage options, valuables to insure, a summary step before submitting.	H	dev
F05	form	Contact form for a wedding photographer; names, event date, venue, package interest, message.	E	dev
F06	form	Patient-intake form for a dental clinic; personal and insurance details, medical-history checkboxes, consent.	M	seen
P01	pricing	Pricing page for a note-taking app; three tiers (free/personal/team), monthly-annual toggle, feature comparison.	E	dev
P02	pricing	Pricing page for a car-wash membership; three plans, what each includes, an FAQ section.	M	dev
P03	pricing	Pricing page for a freelance illustrator; service packages, add-ons, turnaround times, booking link.	M	dev
P04	pricing	Pricing page for a cloud-storage service; usage-based tiers, an interactive cost estimate, enterprise contact option.	H	dev

id	type	brief	d	split
P05	pricing	Pricing page for a yoga studio; drop-in, class packs, unlimited monthly, a short note on each.	E	dev
P06	pricing	Pricing page for a project-management tool; four plans across a long feature matrix with category groupings.	M	seen
PF01	portfolio	Portfolio home for a furniture maker; project grid, short bio, contact details.	E	dev
PF02	portfolio	Portfolio for a film-score composer; reel placeholder, selected credits, a list of instruments and tools used.	M	dev
PF03	portfolio	Portfolio for an architecture studio; three built projects with descriptions, a studio statement.	M	dev
PF04	portfolio	Portfolio for a data journalist; long-form interactive pieces, each with a synopsis and a role note.	H	dev
PF05	portfolio	Portfolio for a tattoo artist; a gallery, studio location, a booking enquiry link.	E	dev
PF06	portfolio	Portfolio for a motion-graphics freelancer; work organized into categories, a short about section.	M	seen
B01	blog	Article page about restoring a vintage typewriter; headings, image placeholders, an author note.	E	dev
B02	blog	Blog index for a gardening site; recent articles by season, tags, a search field.	M	dev
B03	blog	Long-form article on the history of the tea trade; table of contents, pull quotes, footnotes.	M	dev
B04	blog	Magazine-style feature profiling a wildlife photographer; full-width imagery placeholders mixed with narrative columns.	H	dev
B05	blog	Blog post reviewing three noise-cancelling headphones; a verdict box, pros and cons.	E	dev
B06	blog	Recipe article for a sourdough loaf; ingredients, step-by-step method, timing notes.	M	seen
E01	ecommerce	Product page for a stainless-steel water bottle; image gallery, price, size and color options, add-to-cart.	E	dev
E02	ecommerce	Product page for a mechanical keyboard; switch options, specifications, reviews summary, related products.	M	dev
E03	ecommerce	Product listing for a bookstore's staff picks; filters by genre, sort options, a grid of titles.	M	dev
E04	ecommerce	Configurator page for a made-to-order desk; material, dimensions, cable options, live price, order.	H	dev
E05	ecommerce	Product page for single-origin coffee; roast details, tasting notes, grind selection, subscribe option.	E	dev
E06	ecommerce	Product listing for a plant nursery; filtered by light and care level, each card showing name and price.	M	seen
A01	auth	Login screen for a budgeting app; email and password, remember-me, reset-password link.	E	dev
A02	auth	Registration screen for a co-working booking site; email, password-strength meter, terms acceptance.	M	dev
A03	auth	Two-factor screen prompting for a six-digit code; resend option, a help note.	M	dev
A04	auth	Account-recovery flow screen; multiple verification methods, explanation of what happens next.	H	dev
A05	auth	Sign-in for a library catalog; card number and PIN, a guest-access option.	E	dev
A06	auth	Single-sign-on chooser screen; pick an organization before continuing to log in.	M	seen
ST01	settings	Settings page for a chat app; notifications, privacy, appearance, connected devices, using labeled toggles and selects.	M	unseen
ST02	settings	Account settings for a streaming service; profile, subscription, playback quality, parental controls, downloads.	H	unseen
ST03	settings	Workspace settings for a team tool; members, roles and permissions, billing, integrations.	M	unseen
ST04	settings	Notification-preferences page; choose channels (email/push/SMS) per category, a save bar.	E	unseen
ST05	settings	Device settings for a smart thermostat; schedules, temperature units, eco mode, sensor calibration.	H	unseen
AT01	admin-table	Admin users table; name, email, role, status, last-active columns, search, row actions.	M	unseen
AT02	admin-table	Admin orders table; filterable columns, status badges, bulk selection, pagination, detail-drawer trigger.	H	unseen

id	type	brief	d	split
AT03	admin-table	Admin content-moderation queue; reported items with reason, reporter, date, approve/remove actions.	M	unseen
AT04	admin-table	Admin inventory table; SKU, product, stock, price, an edit action per row, a filter bar.	E	unseen
AT05	admin-table	Admin analytics table; campaigns with sortable metric columns, sparkline placeholders, an export control.	H	unseen

C Statistical and interpretability derivations

Notation is identical to the project’s theory appendix; each result names the section it lifts.

C.1 Factorial model, and the necessity/sufficiency identity (T1). Code a cell by $x \in \{\pm 1\}^5$. The effects parameterization of the 2^5 writes the expected response as the multilinear polynomial $\mathbb{E}[y(x)] = \beta_0 + \sum_i \beta_i x_i + \sum_{i < j} \beta_{ij} x_i x_j + \dots$. Evaluating at the relevant corners (only x_i flips; partners held at +1 for necessity, −1 for sufficiency) gives Eq. 1: the main term contributes $2\beta_i$ in both, while a pair $\{i, j\}$ contributes $+2\beta_{ij}$ when the partner is present and $-2\beta_{ij}$ when absent. Hence $\frac{1}{2}(\text{Nec}_i + \text{Suff}_i) = 2\beta_i + R$ and $\frac{1}{4}(\text{Nec}_i - \text{Suff}_i) = \sum_{j \neq i} \beta_{ij}$ (exact through three-factor terms, which are equal in both contrasts and cancel in the difference). Necessity and sufficiency thus coincide iff a component’s interactions vanish; synergy makes a component more necessary than sufficient.

C.2 Resolution-V aliasing (T2). The half-fraction with generator $E=ABCD$ has defining subgroup $\{I, ABCDE\}$; the alias of an effect S is $S \cdot ABCDE$ reduced modulo $x_i^2=1$. The shortest defining word has length 5, so the design is resolution V: main effects alias only with four-factor interactions ($A=BCDE$) and two-factor interactions only with three-factor interactions ($AB=CDE$). Under “three-factor-and-higher negligible,” all five mains and all ten 2FIs are cleanly and simultaneously estimable from 16 runs.

C.3 Mixed model and pseudoreplication (T3). With $y_{cps} = \mu + \tau_c + u_p + v_{cp} + \epsilon_{cps}$, two seeds on a prompt share u_p , so the intraclass correlation is $\rho_{ICC} = (\sigma_p^2 + \sigma_{cp}^2)/(\sigma_p^2 + \sigma_{cp}^2 + \sigma^2)$ and the variance of a cell mean over G prompts $\times m$ seeds is $\frac{\sigma_y^2}{n}[1 + (m-1)\rho_{ICC}]$. The bracket is the design effect; the effective sample size is

$$n_{\text{eff}} = \frac{n}{1 + (m-1)\rho_{ICC}}. \quad (12)$$

With $m=5$ and $\rho_{ICC}=0.3$, DEFF=2.2, so 200 generations count as ≈ 91 independent points—which the MixedLM recovers by partitioning variance. The paired contrast $d_p = \bar{y}_{\text{FULL},p} - \bar{y}_{\text{LOO},p}$ cancels u_p , which is why cells are seed-matched within a prompt and the bootstrap resamples whole prompts.

C.4 Kappa paradox, in exact numbers (T4). General chance-corrected form $(p_o - p_e)/(1 - p_e)$; the coefficients differ only in p_e (Eq. 6). Two raters, 100 binary items, agreement held at $p_o=0.90$. *Skewed* (both-1 = 85, both-0 = 5, disagree = 10): $\pi_1=0.90$, so $p_e=0.82 \Rightarrow \alpha_K=1 - 0.10/0.18=0.444$, while $p_e^\gamma=0.18 \Rightarrow \text{AC1}=(0.90 - 0.18)/0.82=0.878$. *Balanced* ($\pi_1=0.5$): $p_e=p_e^\gamma=0.5 \Rightarrow \alpha_K=\text{AC1}=0.80$. Identical agreement, α_K moved only by the marginal—the justification for the disjunctive gate.

C.5 Style-controlled BT and omitted-variable bias (T5). With $P(i \succ j) = \sigma((\beta_i - \beta_j) + \gamma^\top(s_i - s_j))$, omitting s makes the estimated gap absorb the cell-mean style shift, $\hat{b}_i - \hat{b}_j \rightarrow (\beta_i - \beta_j) + \gamma^\top(\bar{s}_i - \bar{s}_j)$; if the skill produces longer/denser pages, plain BT inflates the design effect by exactly that term. Including s identifies γ from within-cell style variation and subtracts the bias. The mediation caveat: colorfulness is partly causal, so it is reported controlled and uncontrolled and its legitimate effect recovered through the objective channel.

C.6 Paired power (T6). Eq. 7 reproduces $\approx 194, 85, 782$ decisive pairs for 60/40, 65/35, 55/45 (rounded to 200/80/780). Total pairs relate to decisive pairs by $N = N_{\text{dec}}/p_d$; we do not plug the marginal δ with $p_d < 1$ into the two-parameter formula (that mixes a conditional split with a discordance rate). The two-signal rule as error algebra: the necessity/sufficiency *screen* is $A \cup B$ with $\Pr(\text{false} \mid H_0) \leq \alpha_{\text{obj}} + \alpha_{\text{pref}} \lesssim 0.10$; the Stage-2 headline is $E_1 \cap E_2 \cap E_3 \cap E_4$ with joint false-positive rate $\leq \min_k \Pr(E_k \mid H_0)$, far below any single α .

C.7 Objective metrics (T7). WCAG contrast is Eq. 2. Ngo balance $\text{BM} = 1 - \frac{1}{2}(|BM_v| + |BM_h|)$ from summed weight $\times$ distance on each side of the frame axis; equilibrium $\text{EM} = 1 - \frac{1}{2}(|EM_x| + |EM_y|)$ from the center-of-mass offset; alignment regularity $= 1 - n_{\text{distinct edges}}/n_{\text{elements}}$. Hasler colorfulness $M = \sigma_{rgyb} + 0.3 \mu_{rgyb}$ on opponent axes $rg=R-G$, $yb=\frac{1}{2}(R+G)-B$. Modular type-scale adherence is the fraction of adjacent size ratios within $\pm 10\%$ (log space) of the best-fitting canonical ratio. PSI is Eq. 3; POC is Eq. 4.

C.8 Diff-in-means vs. LDA, and the ablation projector (T8). For equal-covariance Gaussians the Bayes-optimal discriminant is $w^* = \Sigma^{-1}(\mu_1 - \mu_0)$; diff-in-means $v = \mu_1 - \mu_0$ equals w^* iff $\Sigma \propto I$. The whitened direction is optimal for reading but amplifies low-variance nuisance directions when *writing*, pushing activations off-distribution; diff-in-means is the on-manifold mean shift and cancels shared nuisance content, so we locate with probes/LDA and steer with diff-in-means. The ablation operator $P = I - \hat{v}\hat{v}^\top$ is an orthogonal projector: idempotent ($P^2 = I - 2\hat{v}\hat{v}^\top + \hat{v}(\hat{v}^\top \hat{v})\hat{v}^\top = P$ using $\hat{v}^\top \hat{v} = 1$) and symmetric ($P^\top = P$); replacing every residual-writing matrix by PW_{out} bakes the ablation into the weights, equivalent to a runtime hook for all writes after the edit.

C.9 Interventions (T9). Steering and ablation are do-operations (Eqs. 10–11); sufficiency uses the NOSYS base so the effect is the direction’s, not a prompt’s. The specificity requirement $\mathbb{E}[Q \mid \text{do}(h_{\ell^*} + = \alpha^* \hat{r}), \text{NOSYS}] \not\asymp \mathbb{E}[Q \mid \text{NOSYS}]$ for random $\hat{r}$ of equal norm ($k=5$) separates a direction effect from a norm effect. A causal effect that holds on *unseen task types* adds transportability; the anti-steerable fraction and dose-response non-monotonicity are reported because a positive mean can hide a heavy tail.

D Judge template (verbatim)

Role. “You are a senior product designer evaluating two rendered web UIs built for the *same brief*. Judge overall design quality for that brief. Ignore which looks longer or more colorful per se; judge whether choices are deliberate and fit the brief.”

Inputs. the brief text; screenshot A; screenshot B (both desktop 1440×900).

Checklist anchors (from the six metric families, to force reliability): color/palette coherence; layout & visual hierarchy; typography; component/pattern quality; restraint (absence of generic “AI-slop” tells); overall fit-to-brief.

Output schema. `{"winner": "A" | "B" | "tie", "confidence": 1-5, "per_criterion": {"color": ..., "layout": ..., "typography": ..., "components": ..., "restraint": ..., "fit": ...}, "rationale": "<=40 words"}`.

Protocol. Each pair is judged as (A,B) and (B,A); a winner counts only if consistent across orders, else tie (Davidson tie term in BT). Judge temperature 0; rubric cached as a system prompt; batched with backoff. Ties feed the tie term, not a half-win/half-loss split.

E Additional figures and tables

Figure 13 is the metric-redundancy diagnostic; the POC z -scores and averages a preregistered subset, not all metrics, so redundancy within a family does not double-count. The localization (F5), stability (F13), PSI-validation (F10), and reliability (F11) figures appear in the main text (Sections 4, 6). Table T10 (planned vs. achieved decisive pairs and BT-utility simulation power per contrast) is generated from the power module on the observed tie rates.

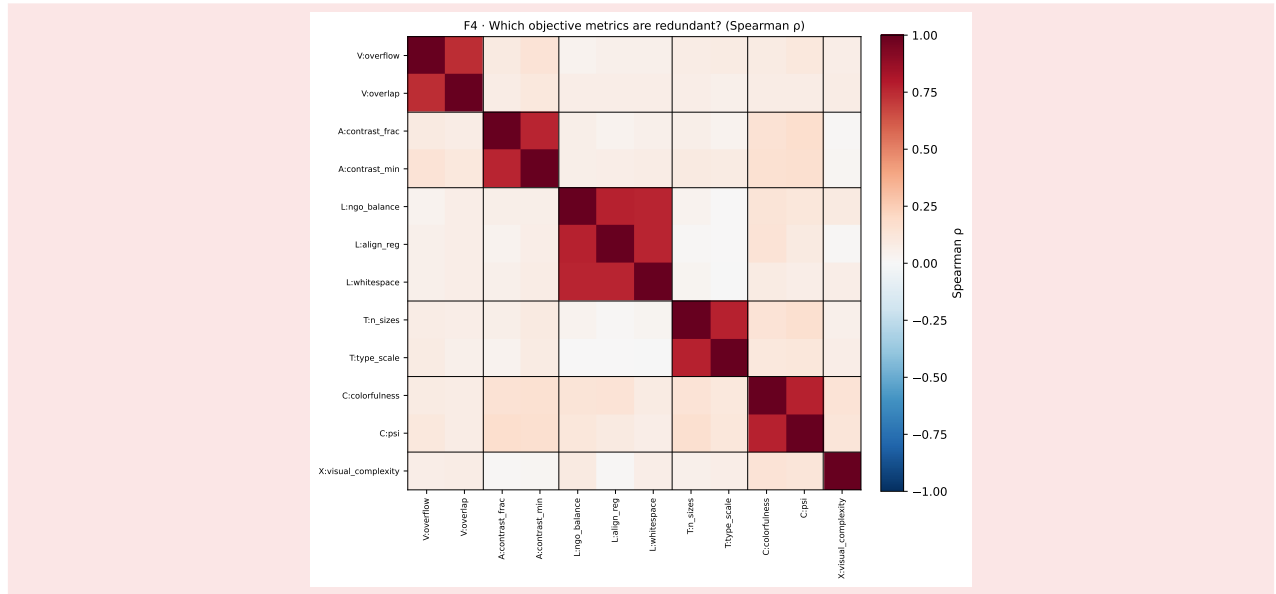


Figure 13: **[F4 – SYNTHETIC PREVIEW.]** *Synthetic pipeline-validation preview; replaced by real data (RUNBOOK session 4).* Metric-correlation matrix (Spearman ρ), blocked by family. Exposes redundancy so the POC’s preregistered subset is defensible; the family-block structure (within-family redundancy, cross-family separation) is what the real 4,630-render matrix shows. Planted block structure here.

F Preregistration summary and amendment log

The confirmatory design is frozen before any GPU run and git-tagged **prereg-v1**. Frozen items: the 24-cell table (Table 2); the 58-brief split by id (Appendix B); the POC formula (Eq. 4) and its two conditionals (PSI-out 6-term variant; c^* fallback); the style-controlled BT endpoint (Eq. 5); the 12-test confirmatory family $\{\text{FULL-NEUTRAL}; 5 \times \text{FULL-LOO-}C_i; 5 \times \text{AOI-}C_i\text{-NEUTRAL}; \text{FULL-BEAUTY1}\}$, Holm at FWER 0.05 separately on POC and BT, with everything else BH at FDR 0.10; the reliability gate ($\alpha_K \geq 0.667$ or $\text{AC1} \geq 0.80$ under demonstrable skew); the power gate (achieved $\geq$ required decisive pairs at the observed split); the Stage-2 success criterion (Section 6); and the null rule (Section 4). The cell table, confirmatory family, success criterion, and null rule may not be weakened post-freeze; a confirmatory change forks **prereg-v2** and is reported as a deviation.

Amendment log. (*none — freeze pending the **prereg-v1** tag*). Post-freeze amendments will be appended here, dated and numbered, each stating what changed, why, and whether it touches a confirmatory decision.

G Data-science lifecycle map

The paper doubles as the report on each lifecycle phase; Table 7 maps section to phase, to the notebook that carries the narrative, and to the runbook session that produces the real numbers.

Table 7: Paper section $\leftrightarrow$ lifecycle phase $\leftrightarrow$ notebook $\leftrightarrow$ runbook session.

paper section	lifecycle phase	notebook	runbook session
§3 study design	problem framing + data	00–01	— (frozen pre-GPU)
§4 oracle	measurement	02–03	4–5 (metrics, judging)
§5 Stage-1 results	experiments + evaluation	04–05	1–3 (pilot, generation)
§6 Stage-2 method	modeling	06a/06b, 07a	6–7 (extract, sweep, freeze)
§7 Stage-2 results	evaluation	07a/07b	8–9 (verify, stretch)
§8–10	communication + reproducibility	08	— (write-up)